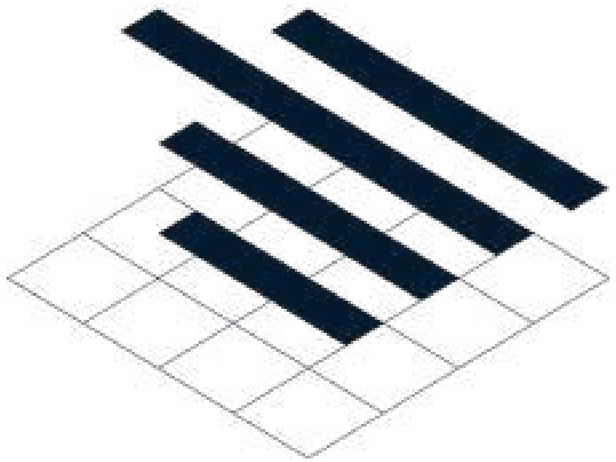
Basic plots with Matplotlib

INTERMEDIATE PYTHON

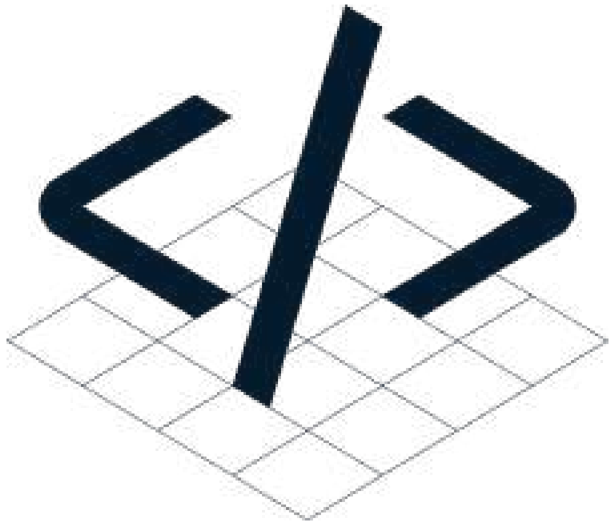
Hugo Bowne-Anderson
Data Scientist at DataCamp

Basic plots with Matplotlib

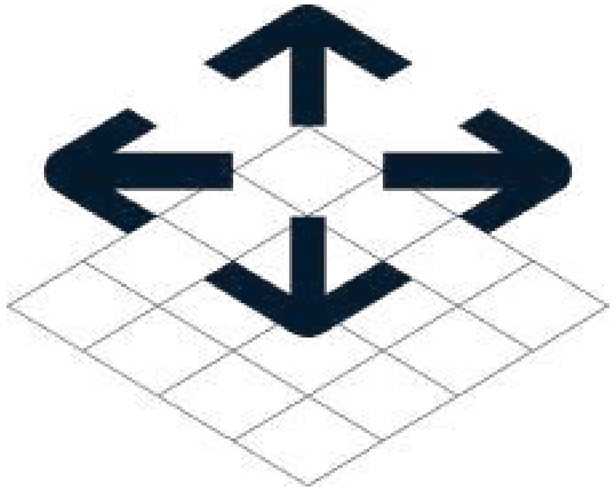
- Visualization



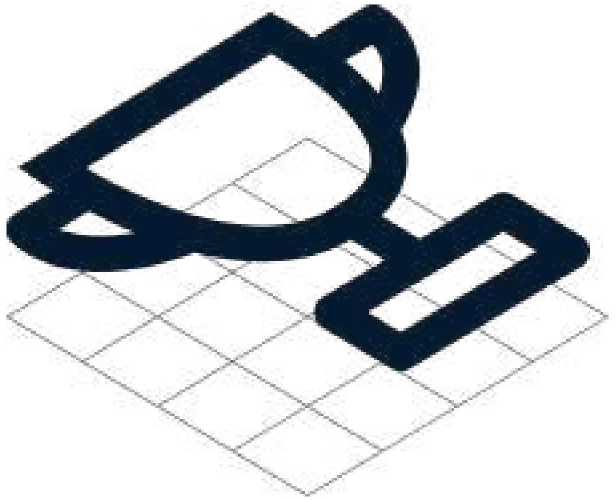
- Data Structure



- Control Structures

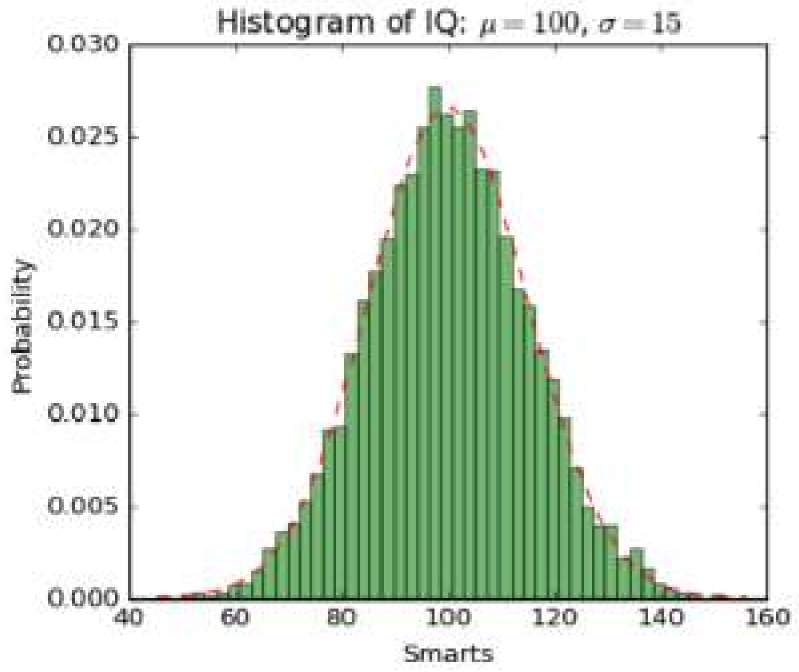


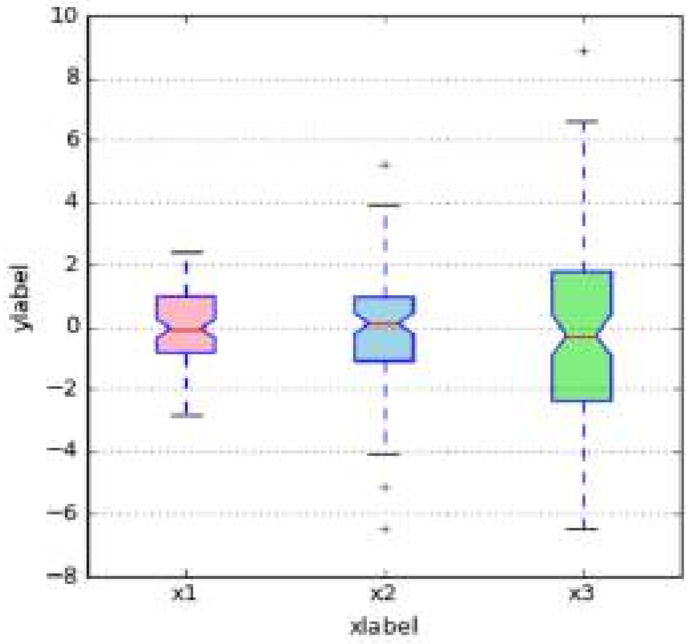
- Case Study

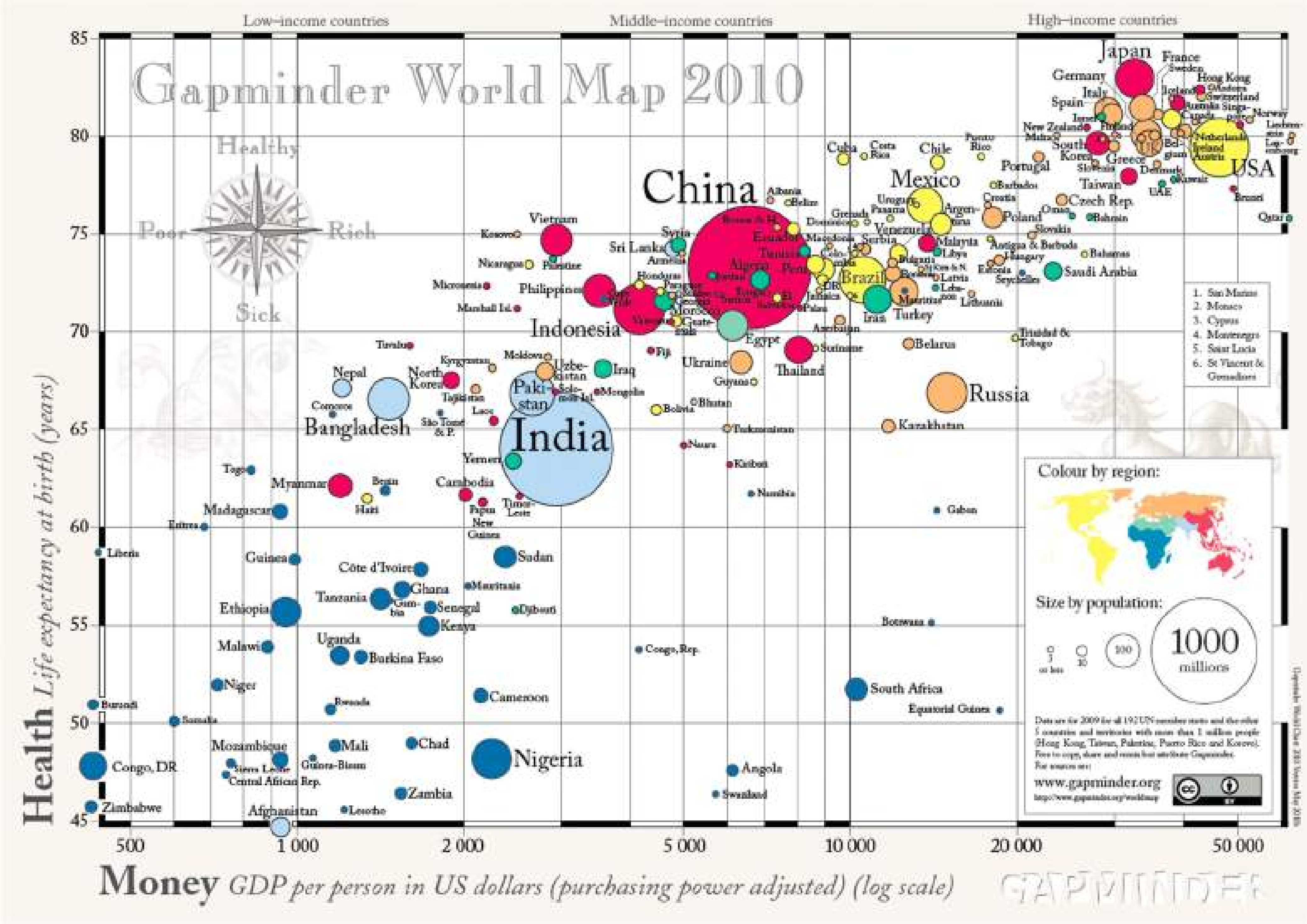


Data visualization

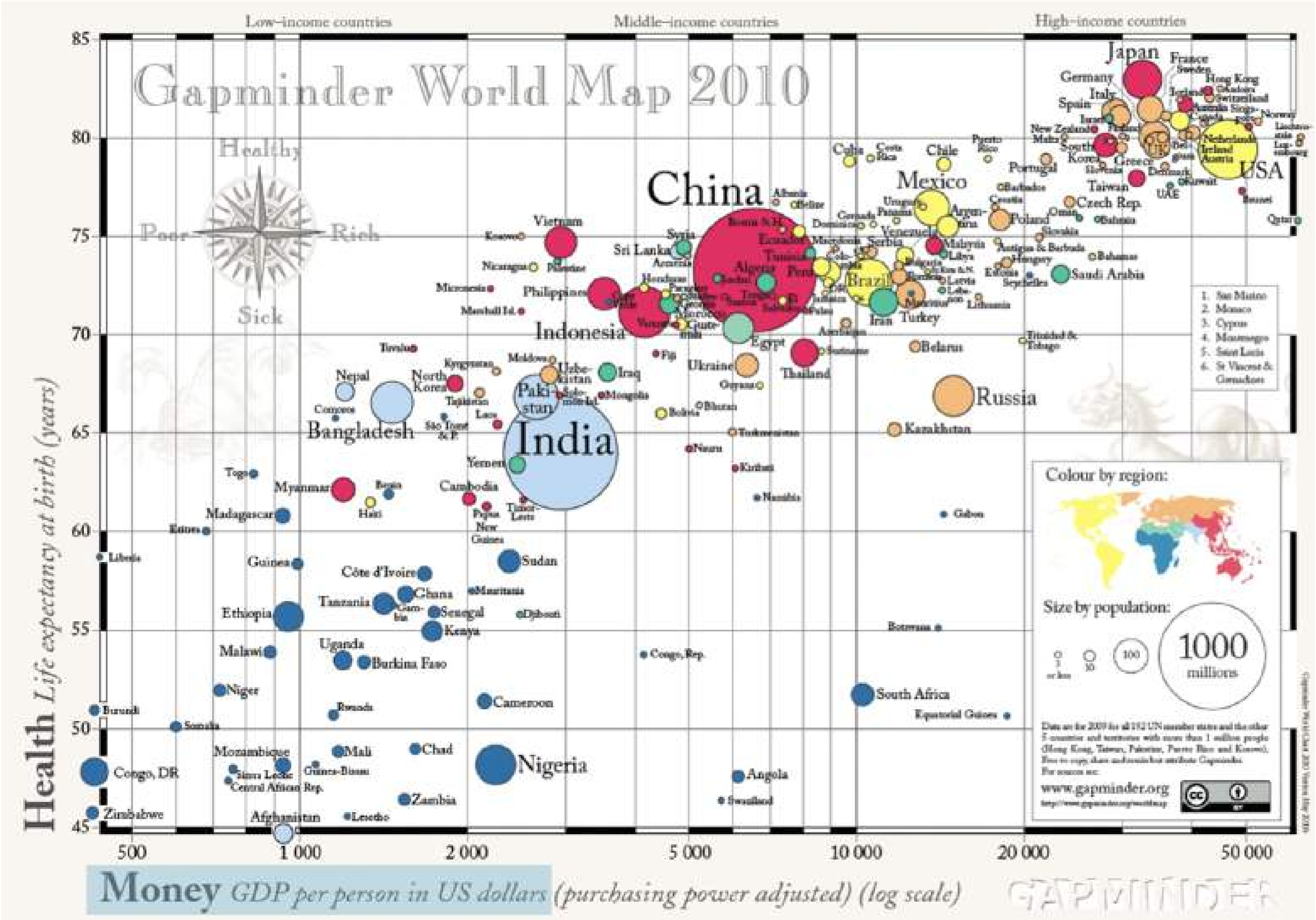
- Very important in Data Analysis
  - Explore data
  - Report insights







[1] Source: GapMinder, Wealth and Health of Nations



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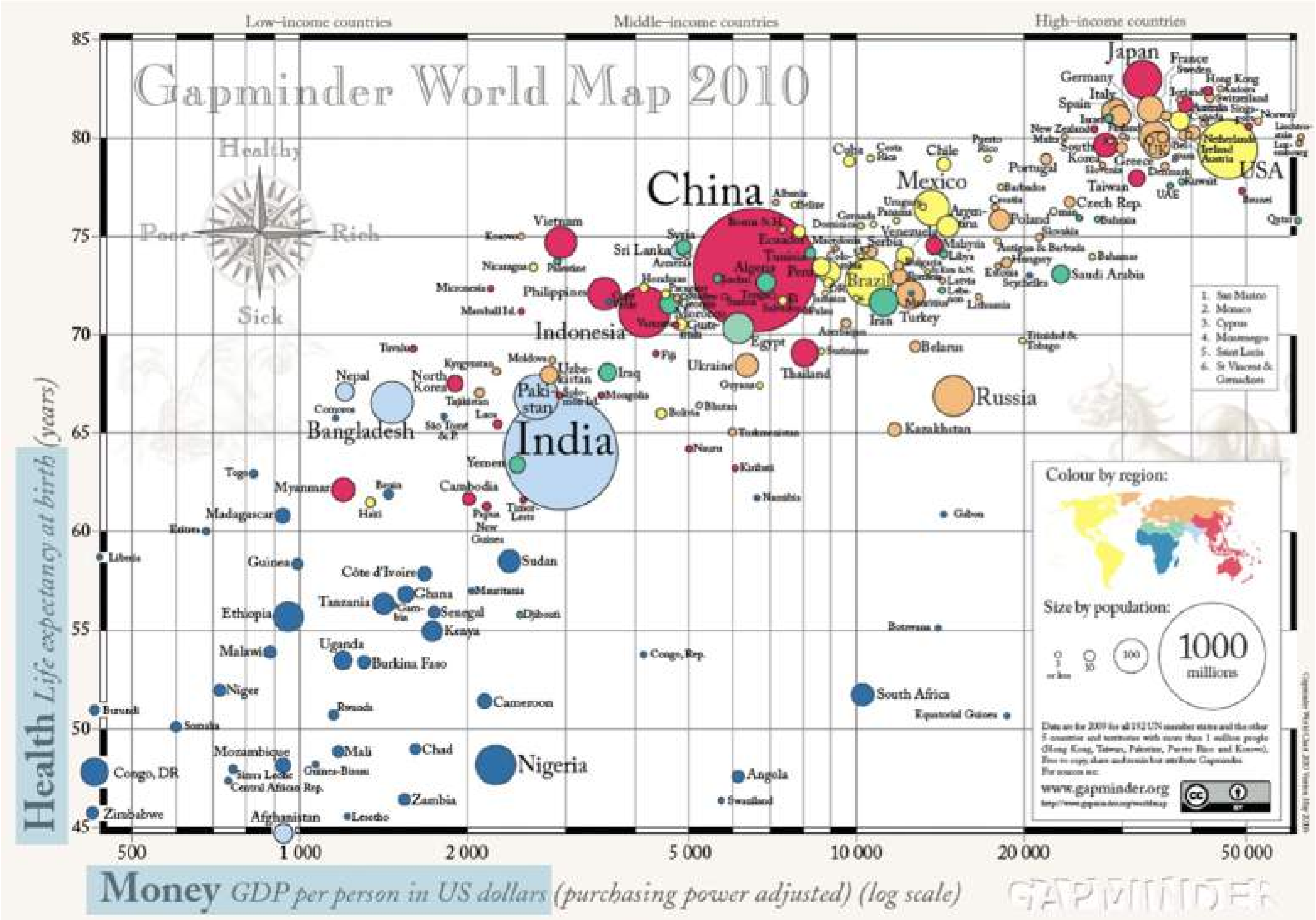
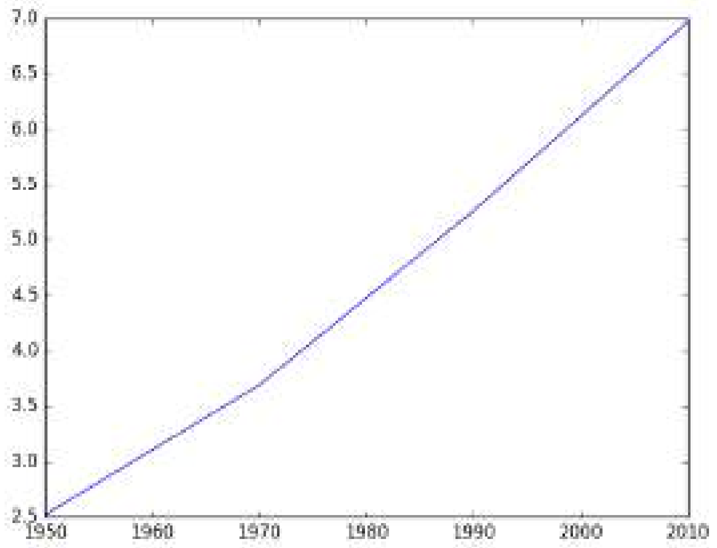


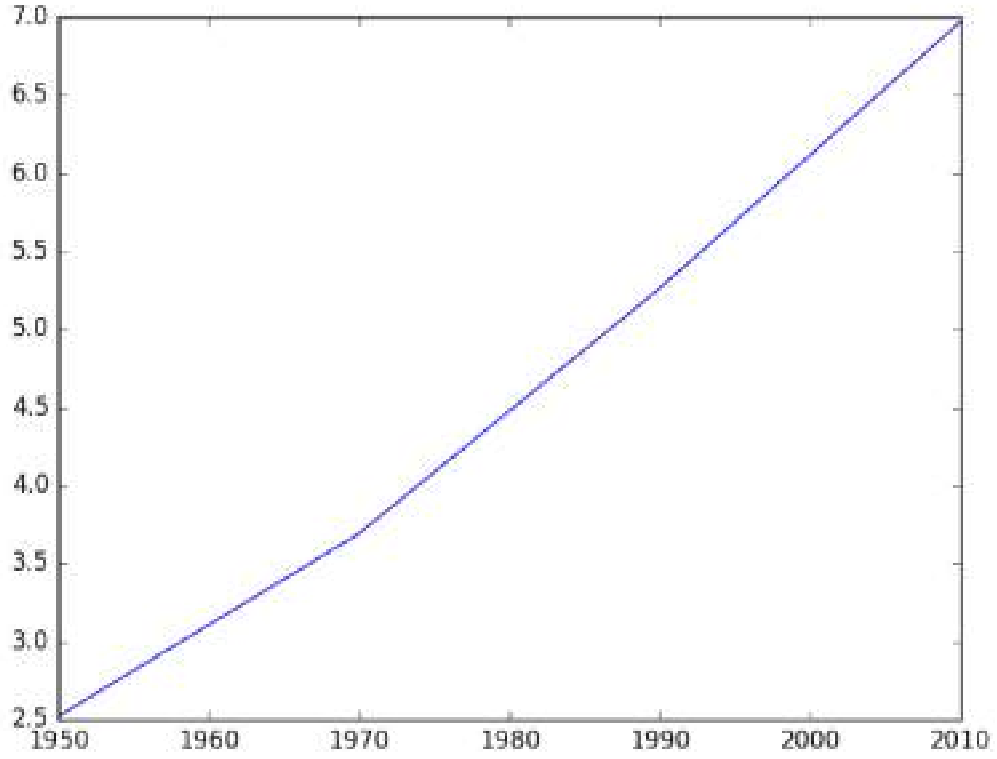
[1] Source: GapMinder, Wealth and Health of Nations

Matplotlib

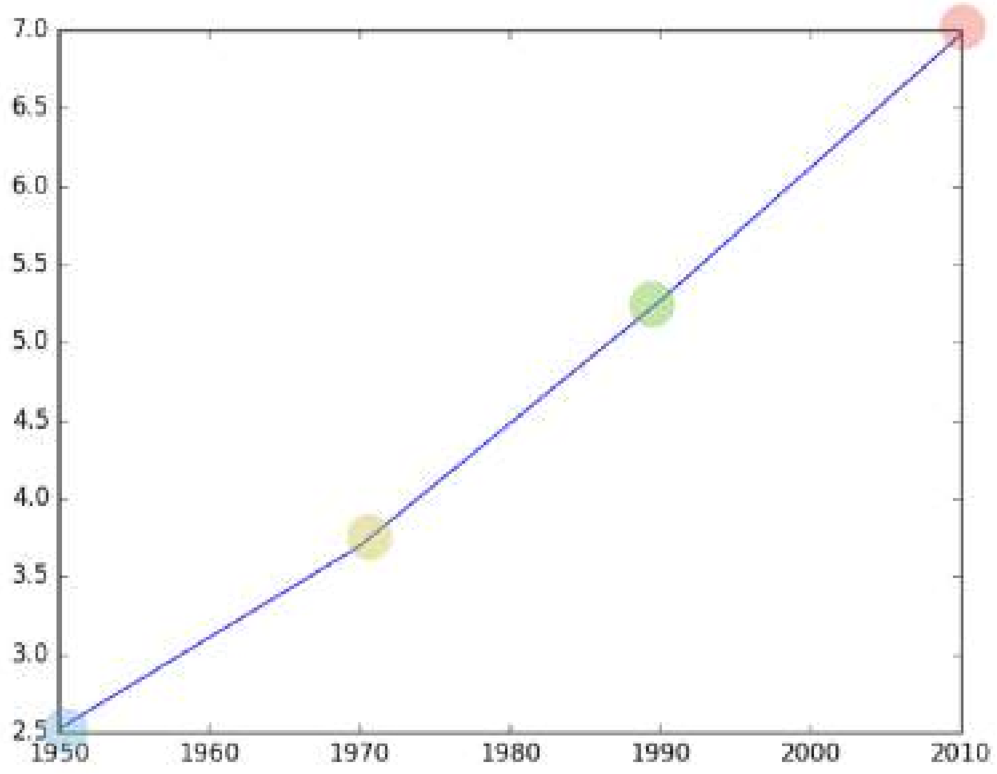
```
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010]
pop = [2.519, 3.692, 5.263, 6.972]
plt.plot(year, pop)
plt.show()
```



Matplotlib



Matplotlib



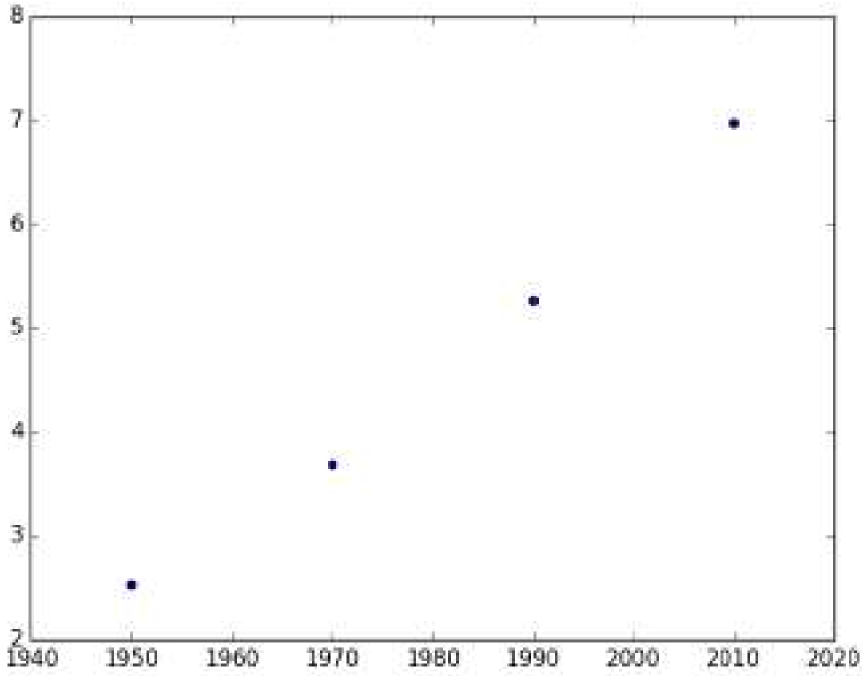

```
year = [1950 , 1970 , 1990 ,  2010]
pop  = [2.519, 3.692, 5.263, 6.972]
```

Scatter plot

```
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010]
pop = [2.519, 3.692, 5.263, 6.972]
plt.plot(year, pop)
plt.show()
```

Scatter plot

```
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010]
pop = [2.519, 3.692, 5.263, 6.972]
plt.scatter(year, pop)
plt.show()
```



Let's practice!

INTERMEDIATE PYTHON

Histogram

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Data Scientist at DataCamp

Histogram

- Explore dataset
- Get idea about distribution

Histogram

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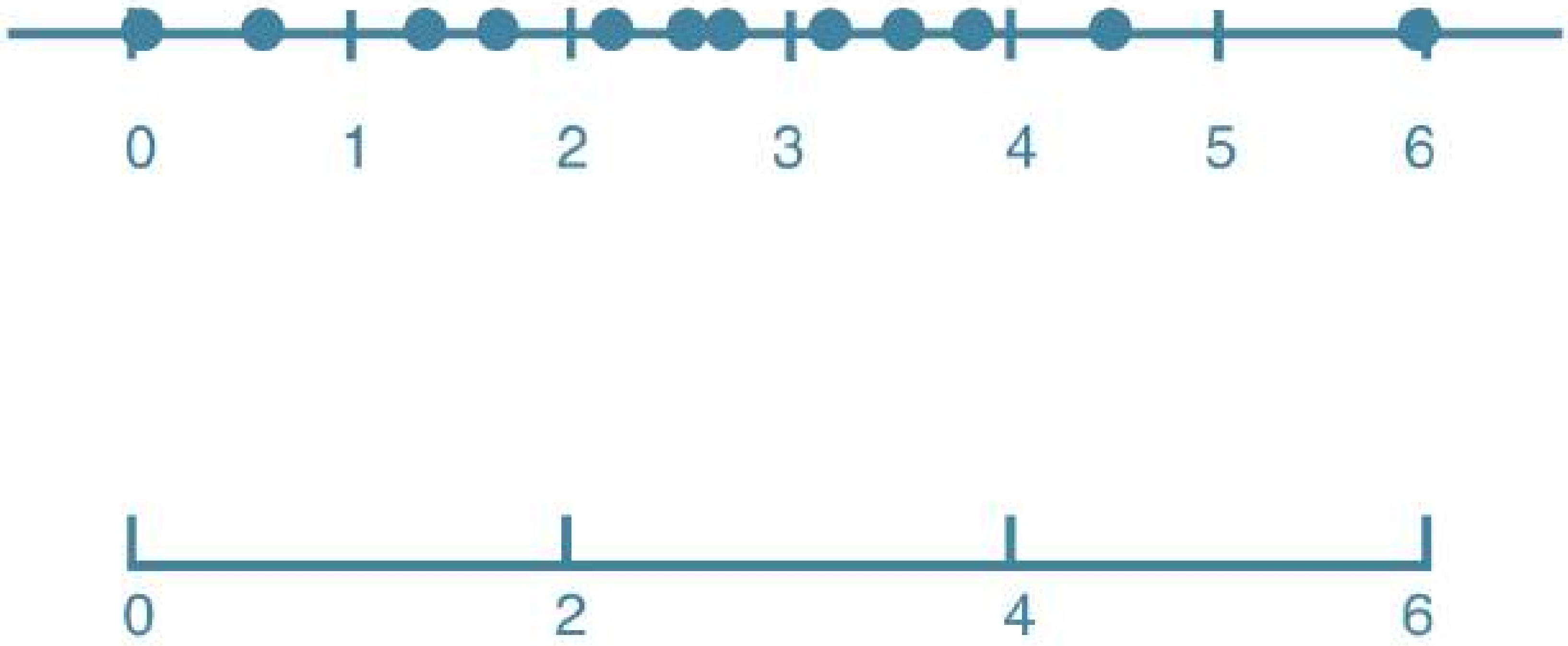

Histogram

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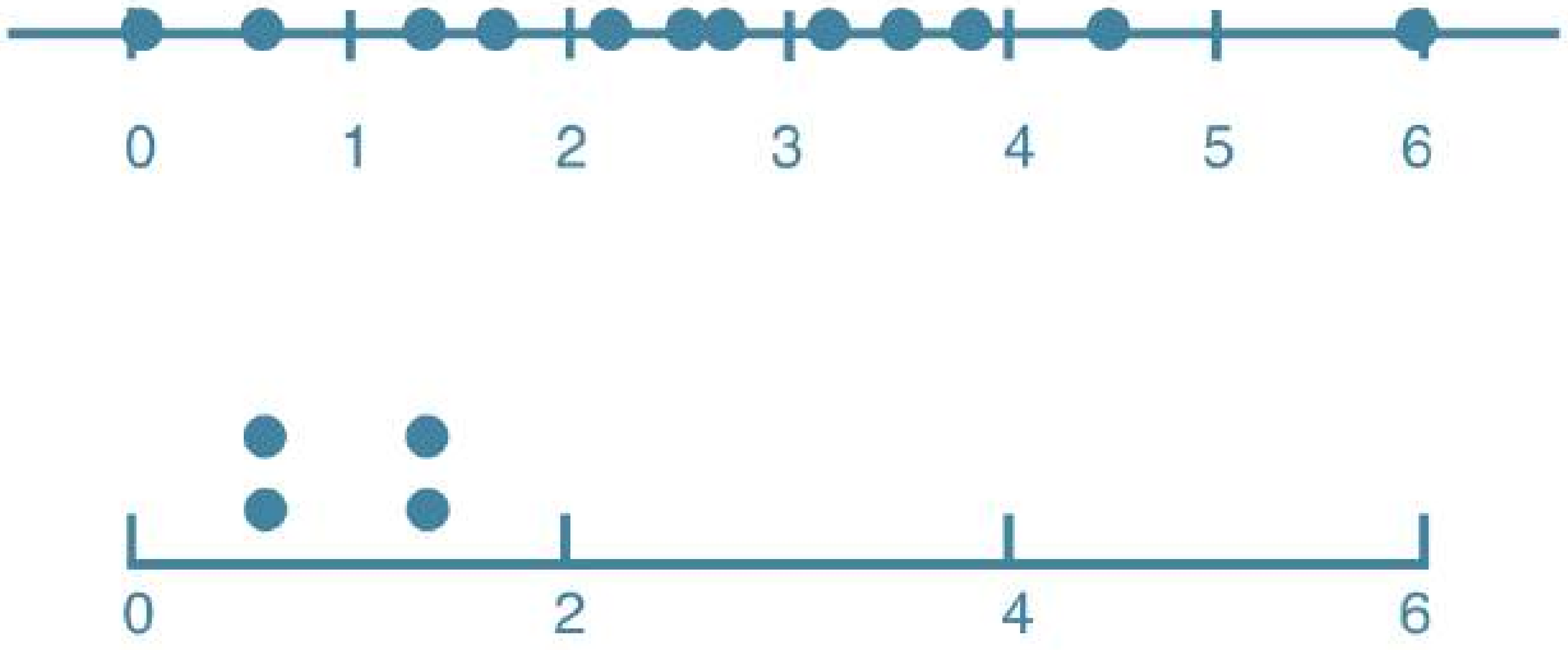
- Get idea about distribution



Histogram

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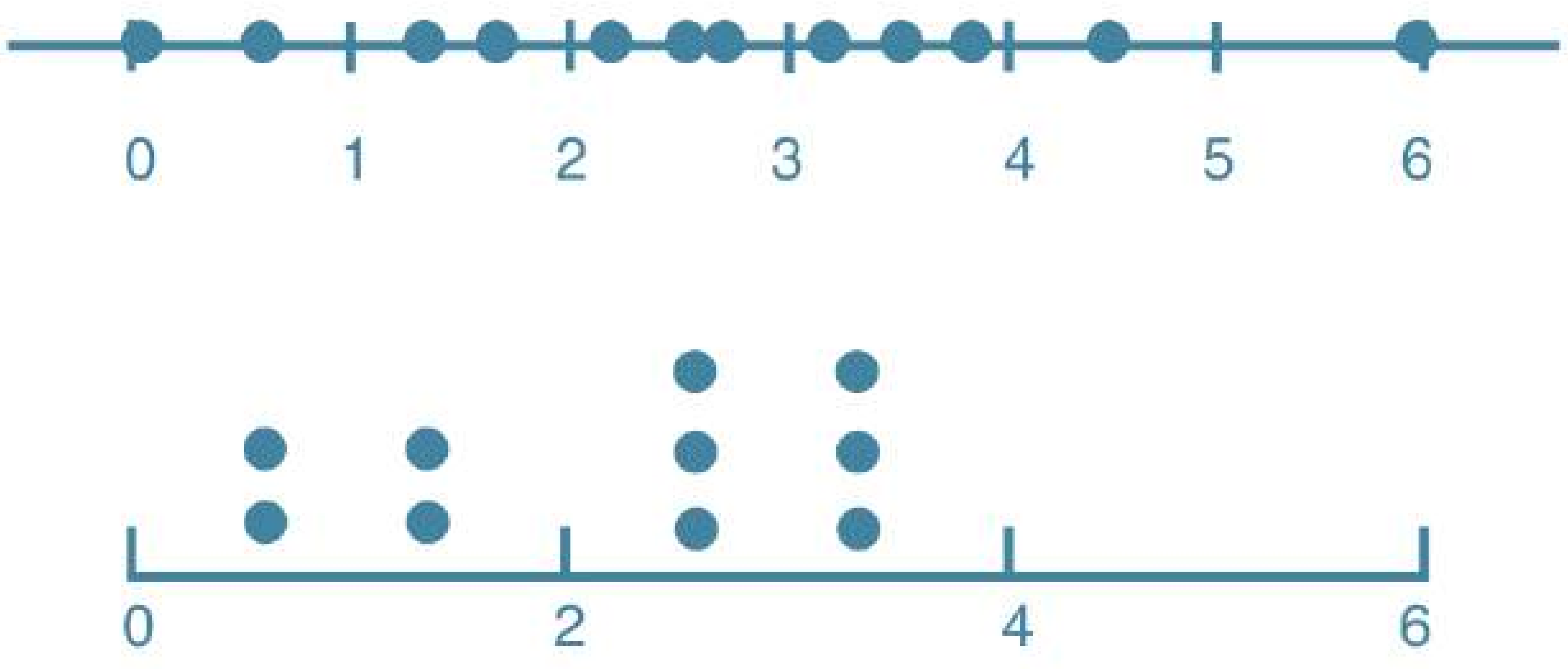
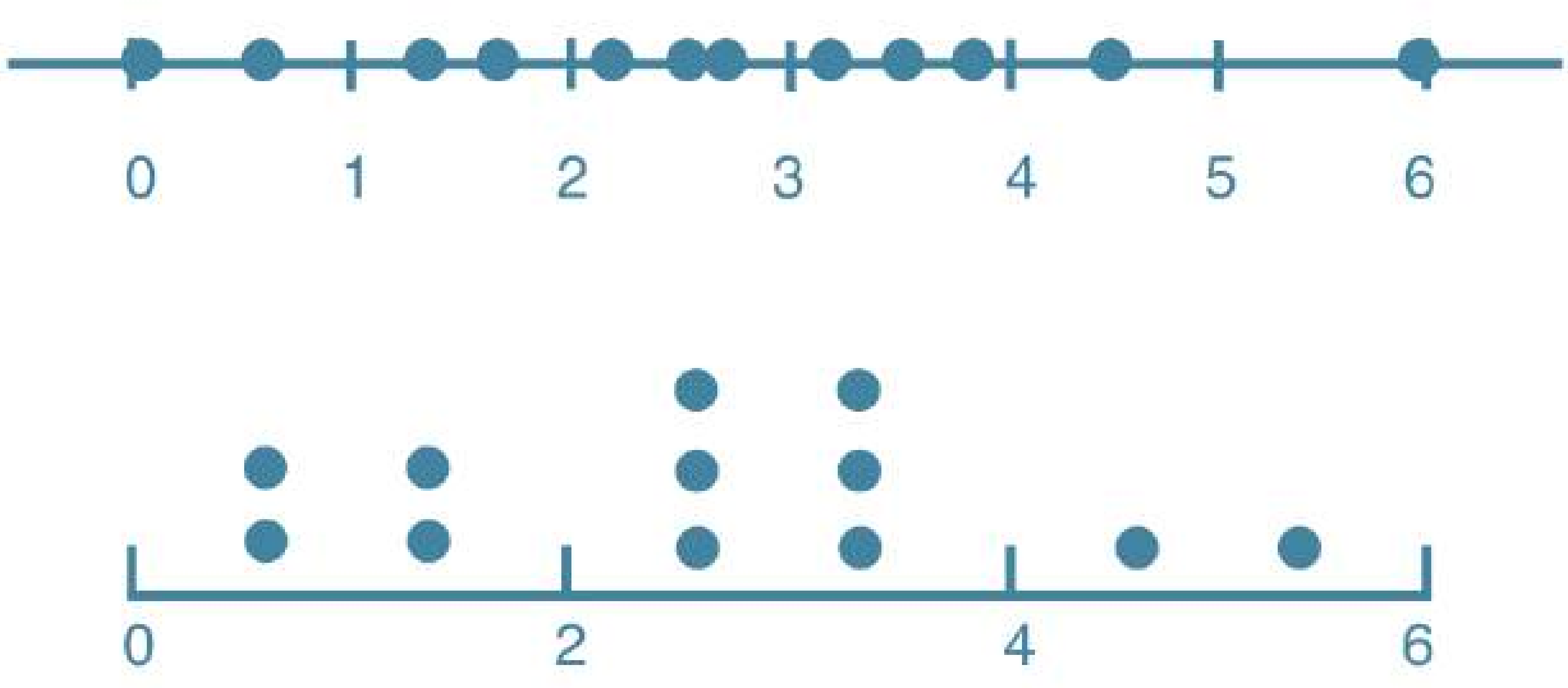


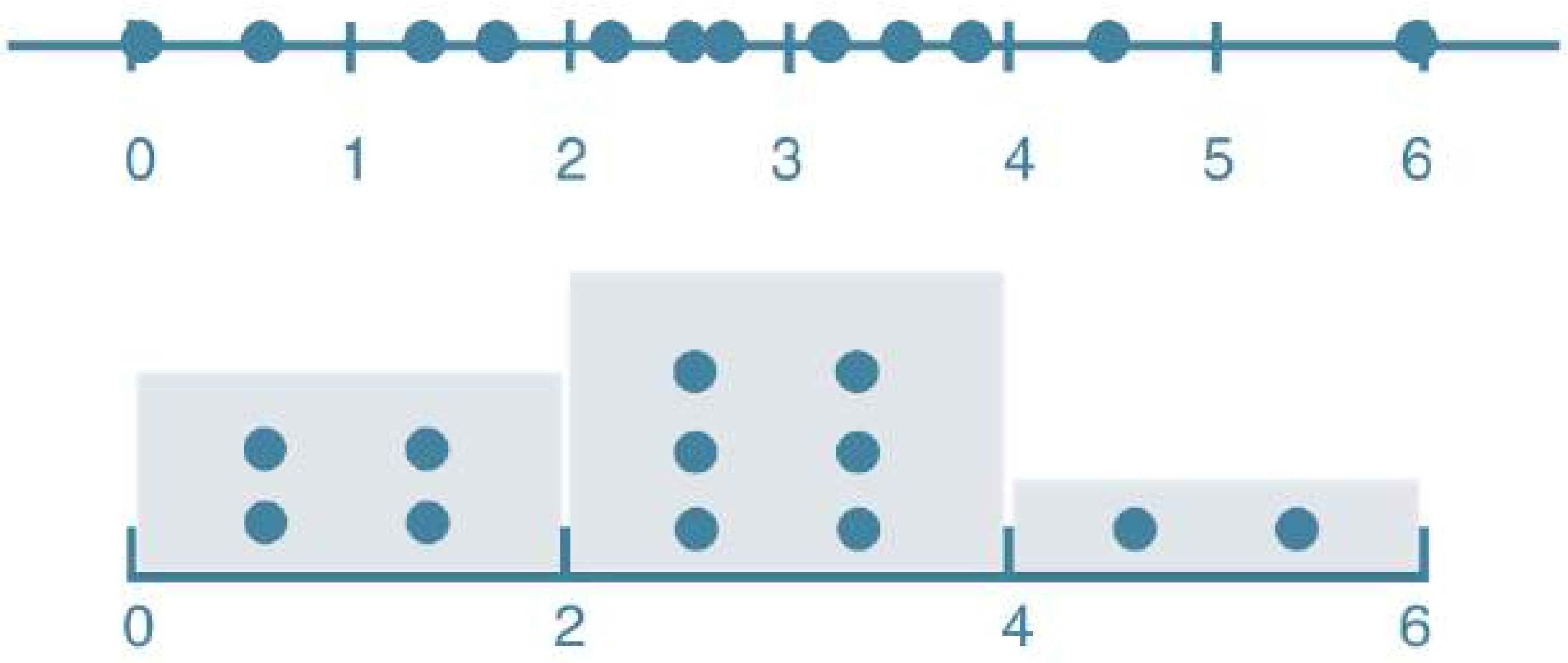
Histogram

- Explore dataset
- Get idea about distribution



Histogram

- Explore dataset
- Get idea about distribution



Matplotlib

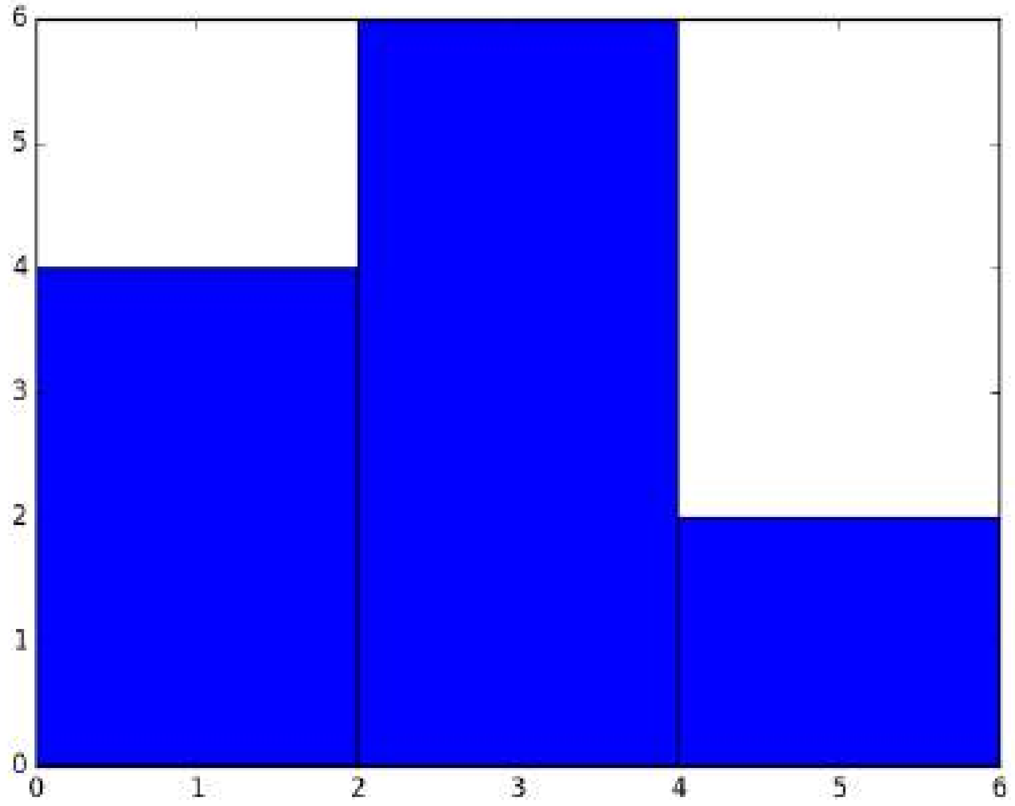
```python
import matplotlib.pyplot as plt
```

```python
help(plt.hist)
```

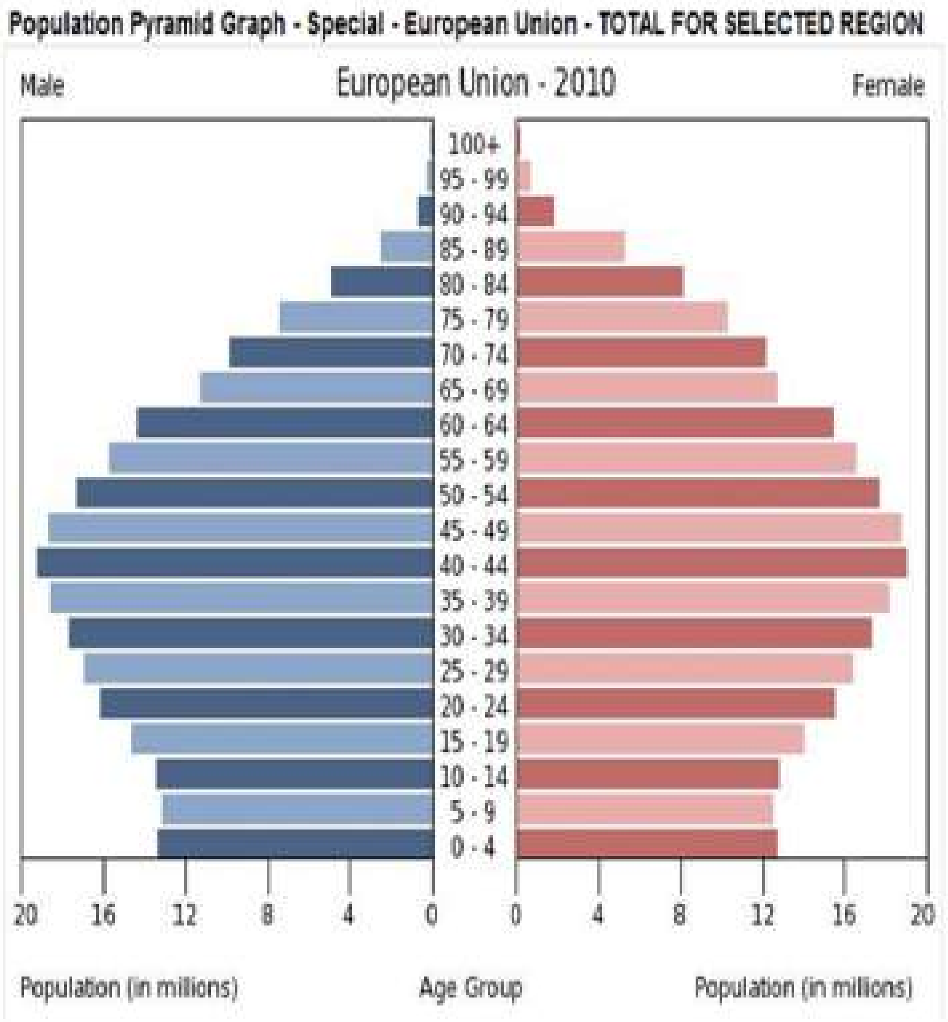
```
Help on function hist in module matplotlib.pyplot:
hist(x, bins=None, range=None, density=False, weights=None,
cumulative=False, bottom=None, histtype='bar', align='mid',
orientation='vertical', rwidth=None, log=False, color=None,
label=None, stacked=False, *, data=None, **kwargs)
    Plot a histogram.

    Compute and draw the histogram of *x*.  The return value is a
    tuple (*n*, *bins*, *patches*) or ([*n0*, *n1*, ...],
    *bins*, [*patches0*, *patches1*, ...]) if the input contains
    multiple data.
```

Matplotlib example

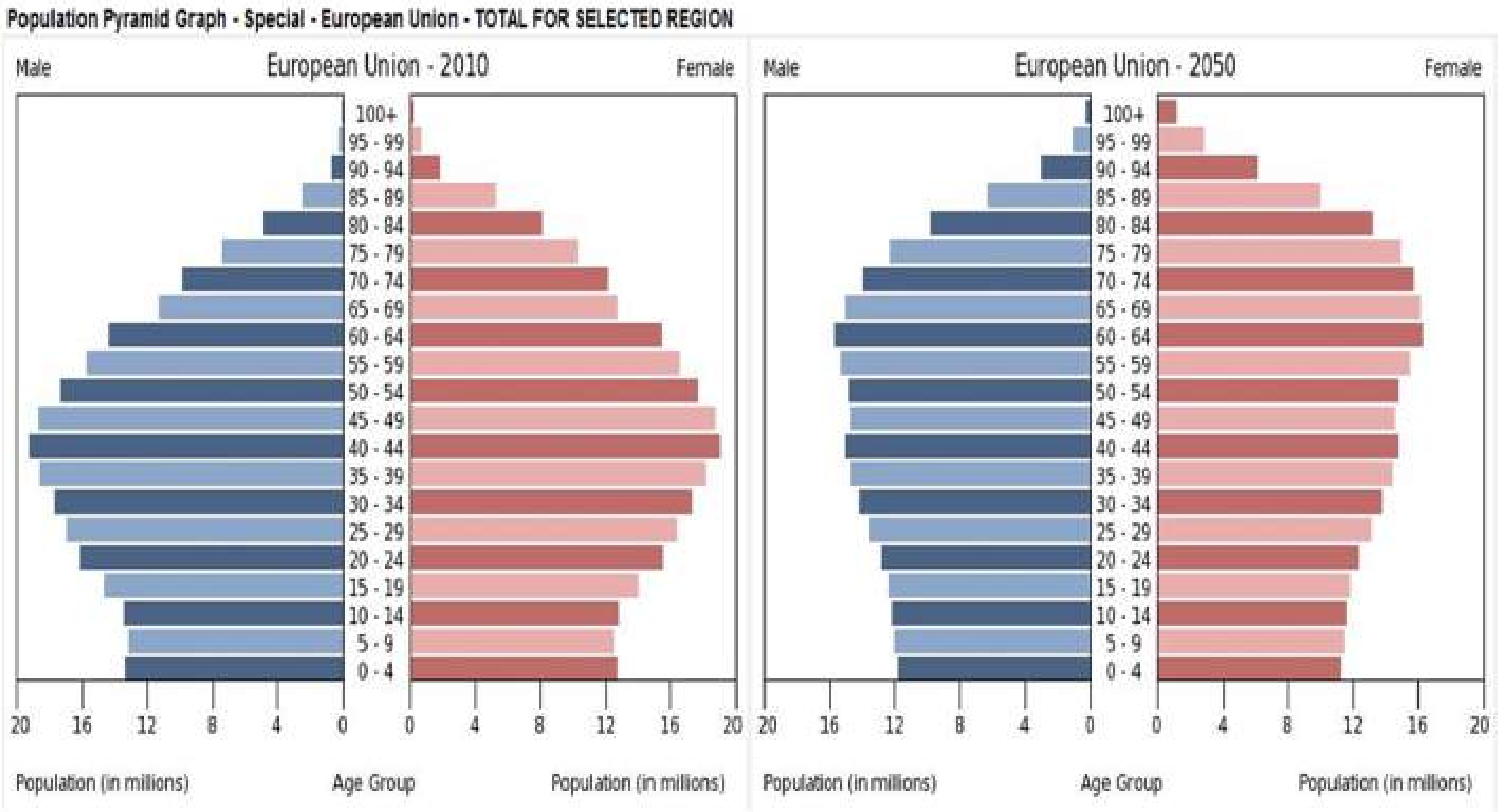
```
values = [0,0.6,1.4,1.6,2.2,2.5,2.6,3.2,3.5,3.9,4.2,6]
plt.hist(values, bins=3)
plt.show()
```



Population pyramid



Population pyramid



Let's practice!

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Customization

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Hugo Bowne-Anderson
Data Scientist at DataCamp

Data visualization

- Many options
  - Different plot types
  - Many customizations
- Choice depends on
  - Data
  - Story you want to tell

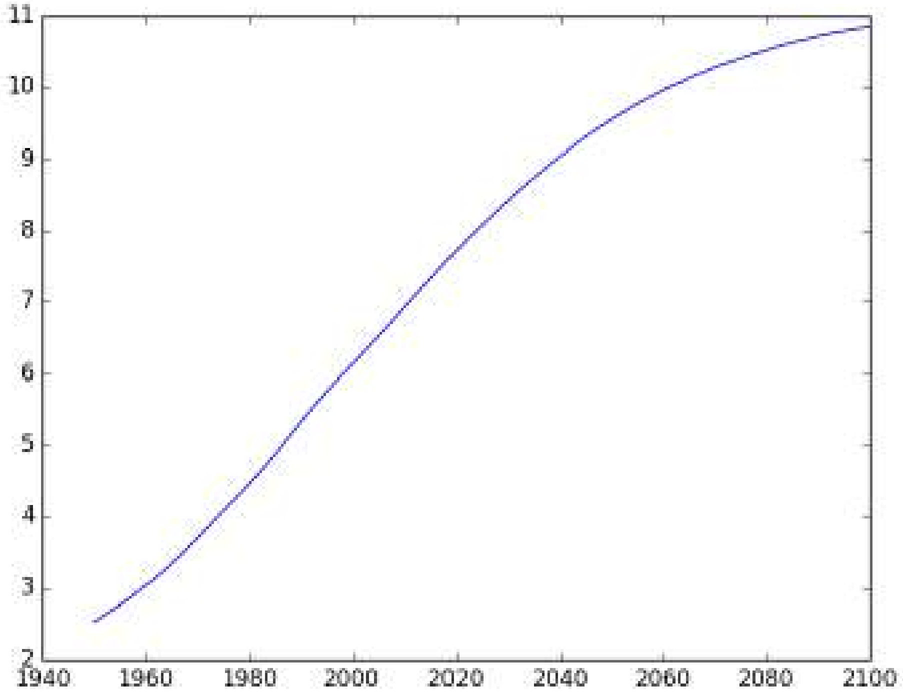
Basic plot

`population.py`

```
import matplotlib.pyplot as plt
year = [1950, 1951, 1952, ..., 2100]
pop = [2.538, 2.57, 2.62, ..., 10.85]

plt.plot(year, pop)

plt.show()
```



Axis labels

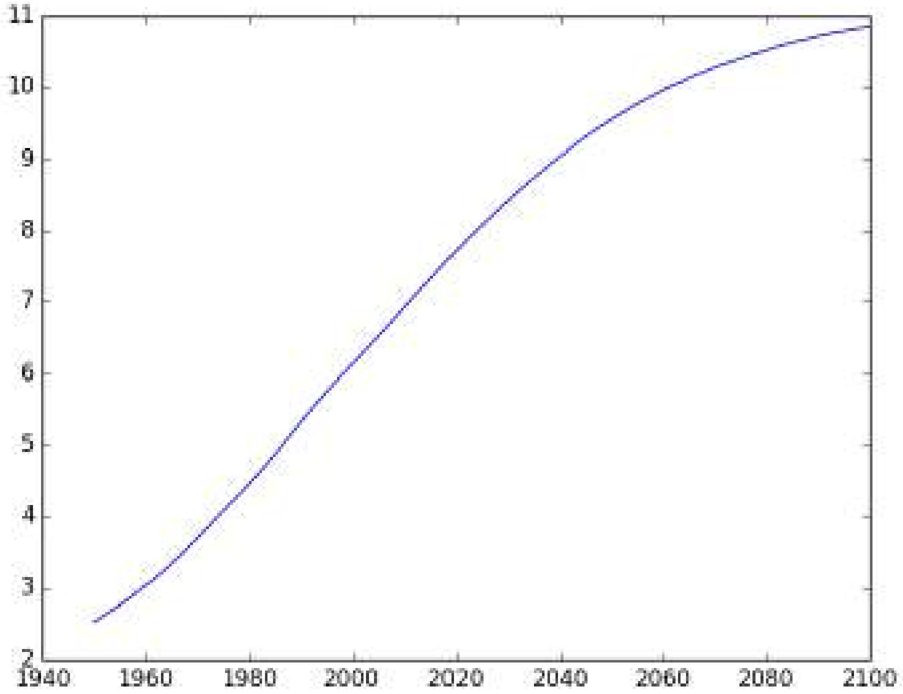
`population.py`

```
import matplotlib.pyplot as plt
year = [1950, 1951, 1952, ..., 2100]
pop = [2.538, 2.57, 2.62, ..., 10.85]

plt.plot(year, pop)

plt.xlabel('Year')
plt.ylabel('Population')

plt.show()
```



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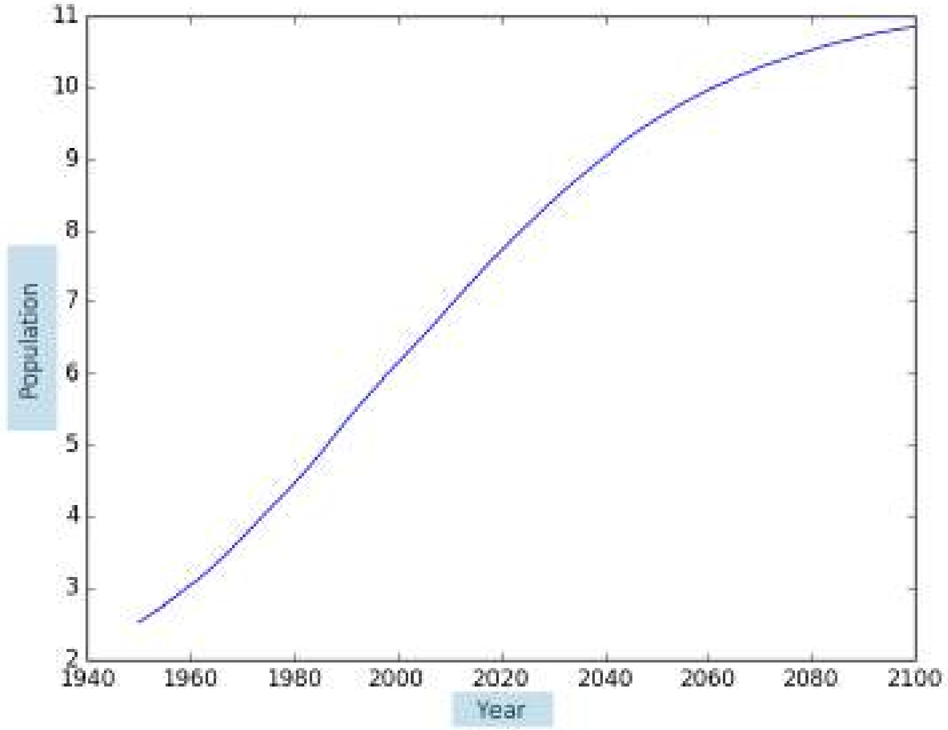


Title

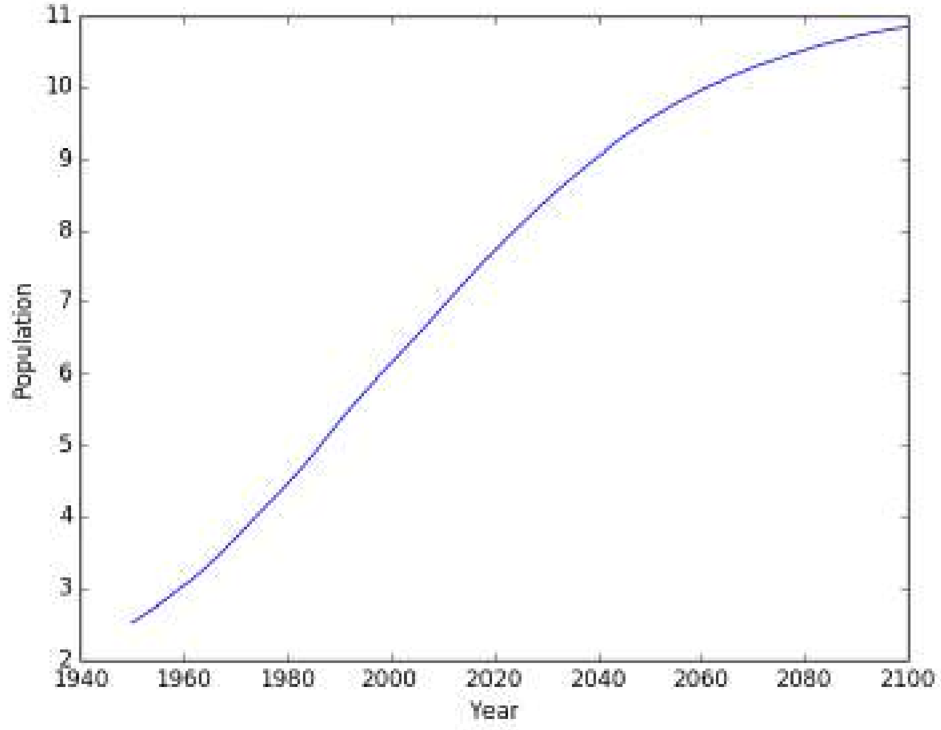
`population.py`

```
import matplotlib.pyplot as plt
year = [1950, 1951, 1952, ..., 2100]
pop = [2.538, 2.57, 2.62, ..., 10.85]

plt.plot(year, pop)

plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')

plt.show()
```



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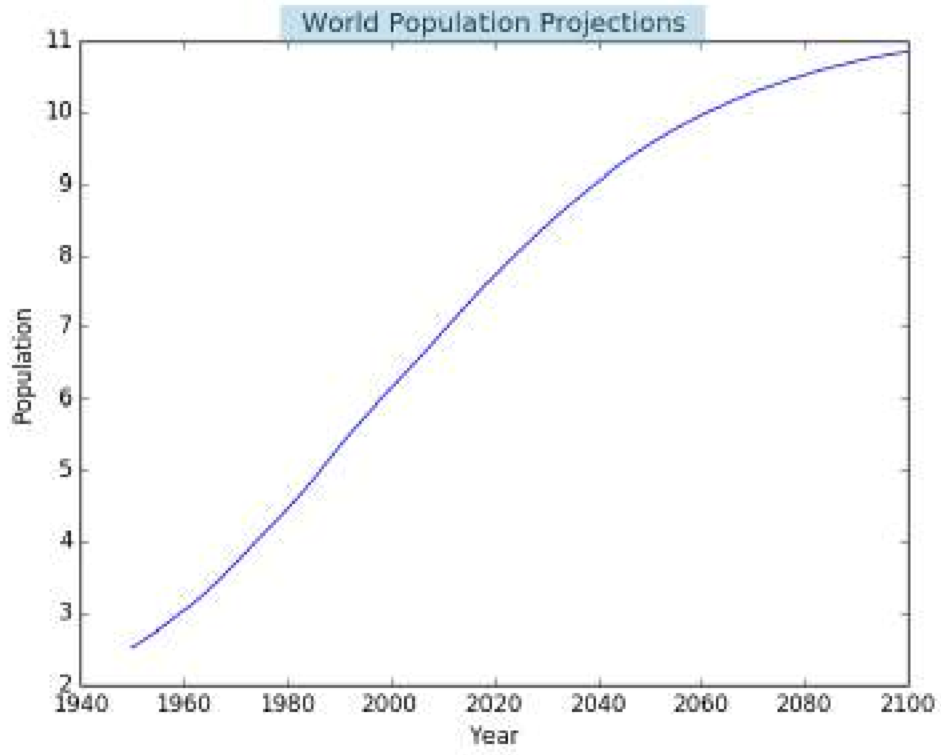


Ticks

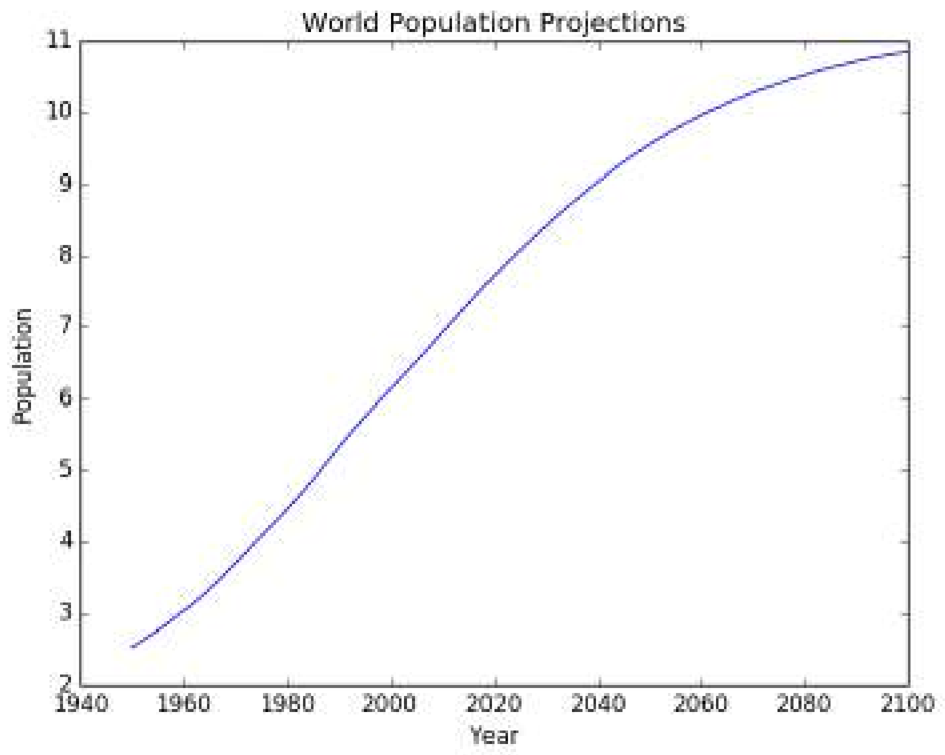
`population.py`

```
import matplotlib.pyplot as plt
year = [1950, 1951, 1952, ..., 2100]
pop = [2.538, 2.57, 2.62, ..., 10.85]

plt.plot(year, pop)

plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10])

plt.show()
```



Ticks

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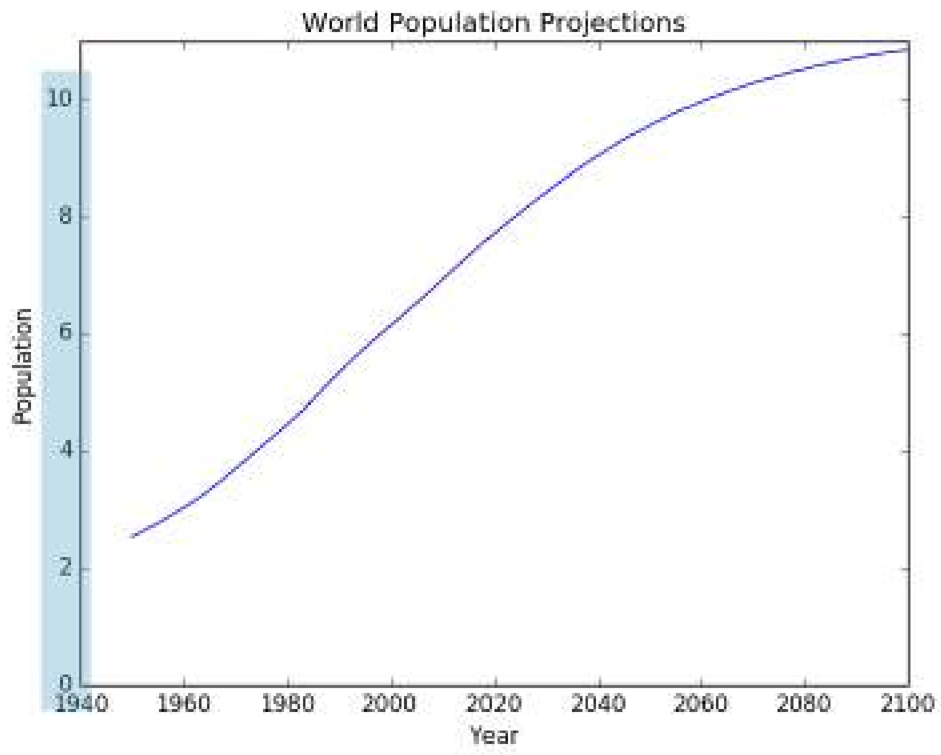
plt.show()
```



Ticks (2)

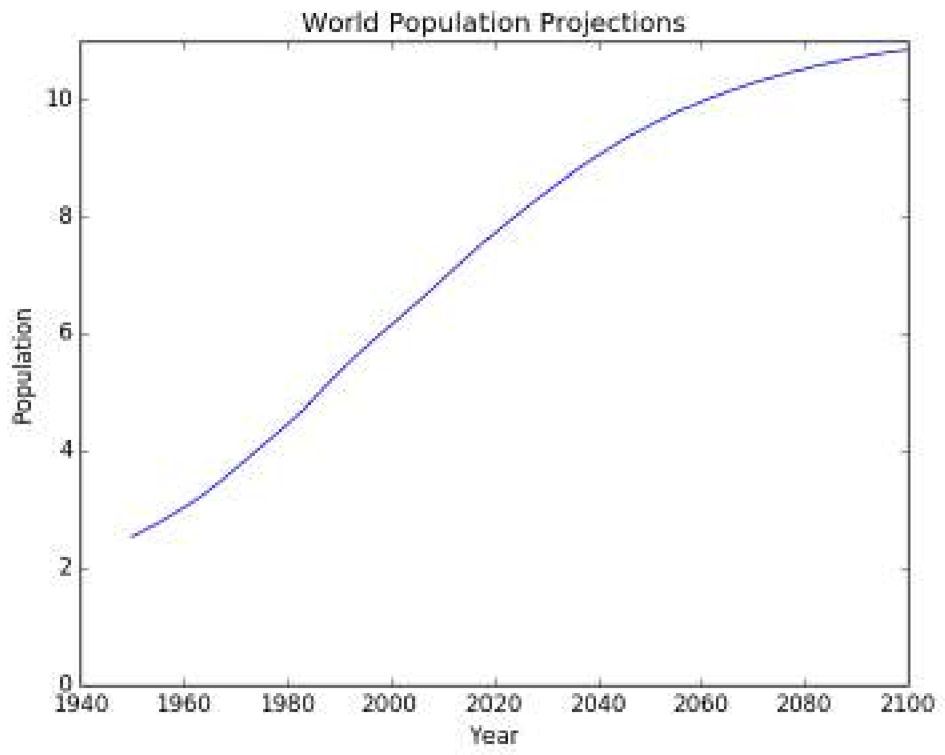
population.py

```
import matplotlib.pyplot as plt
year = [1950, 1951, 1952, ..., 2100]
pop = [2.538, 2.57, 2.62, ..., 10.85]

plt.plot(year, pop)

plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10],
           ['0', '2B', '4B', '6B', '8B', '10B'])

plt.show()
```



Ticks (2)

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plt.show()
```
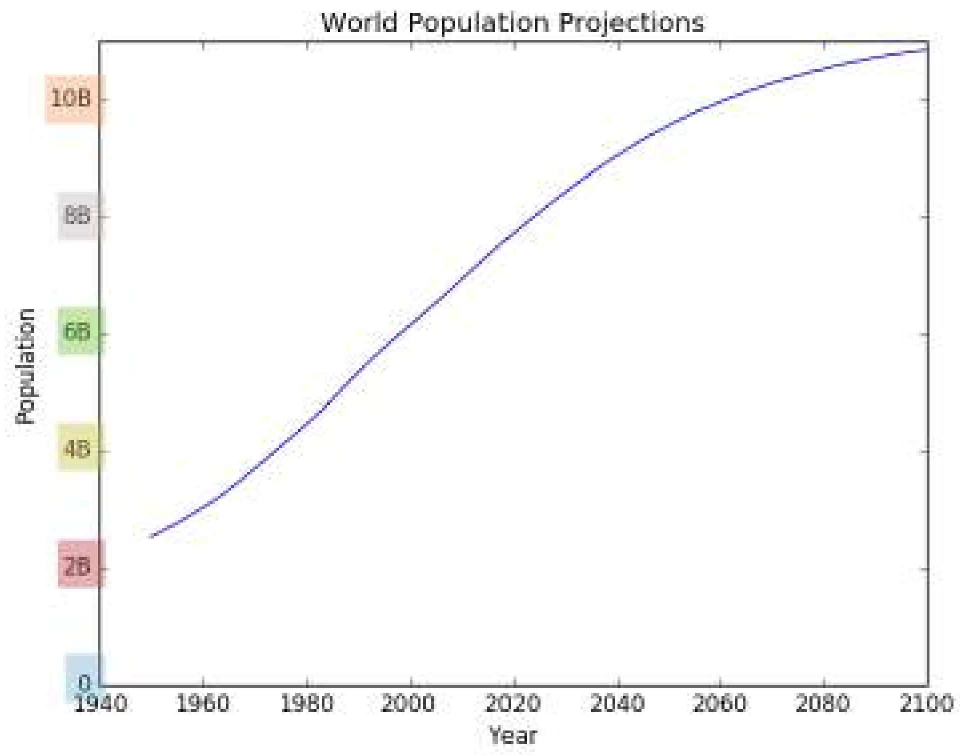



Add historical data

`population.py`

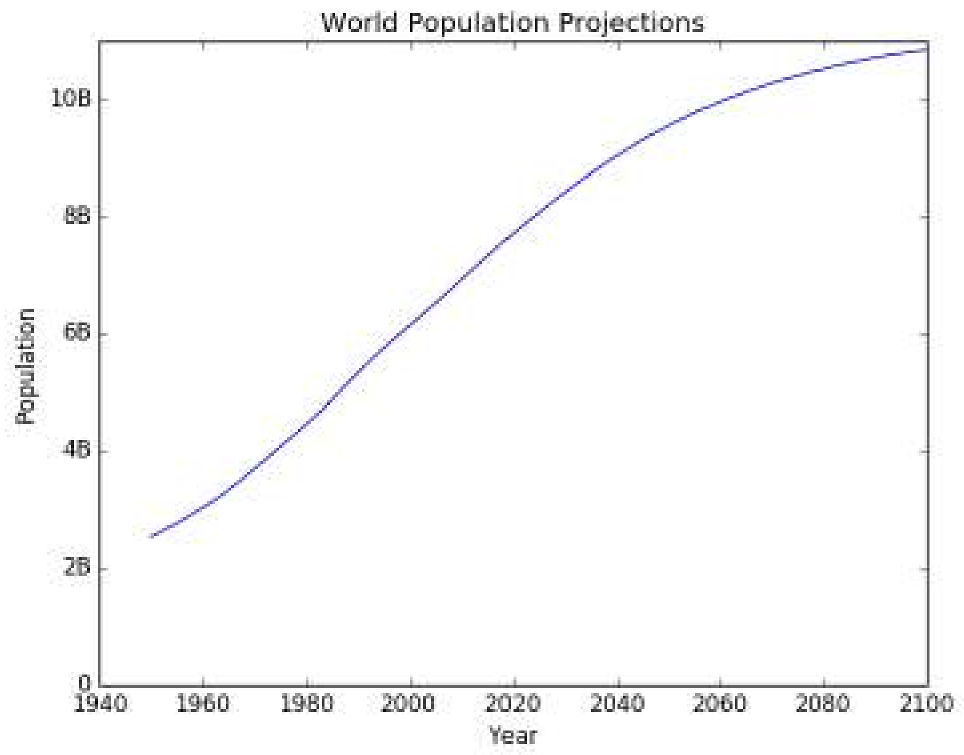
```python
import matplotlib.pyplot as plt
year = [1950, 1951, 1952, ..., 2100]
pop = [2.538, 2.57, 2.62, ..., 10.85]

# Add more data
year = [1800, 1850, 1900] + year
pop = [1.0, 1.262, 1.650] + pop

plt.plot(year, pop)

plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10],
           ['0', '2B', '4B', '6B', '8B', '10B'])

plt.show()
```



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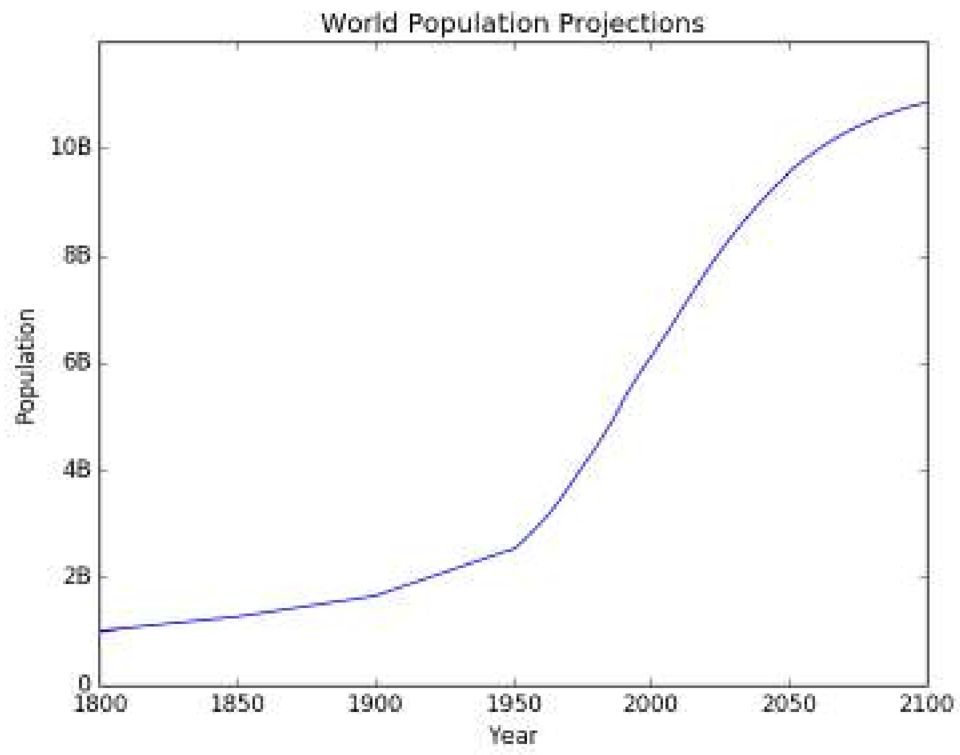
pop = [1.0, 1.262, 1.650] + pop

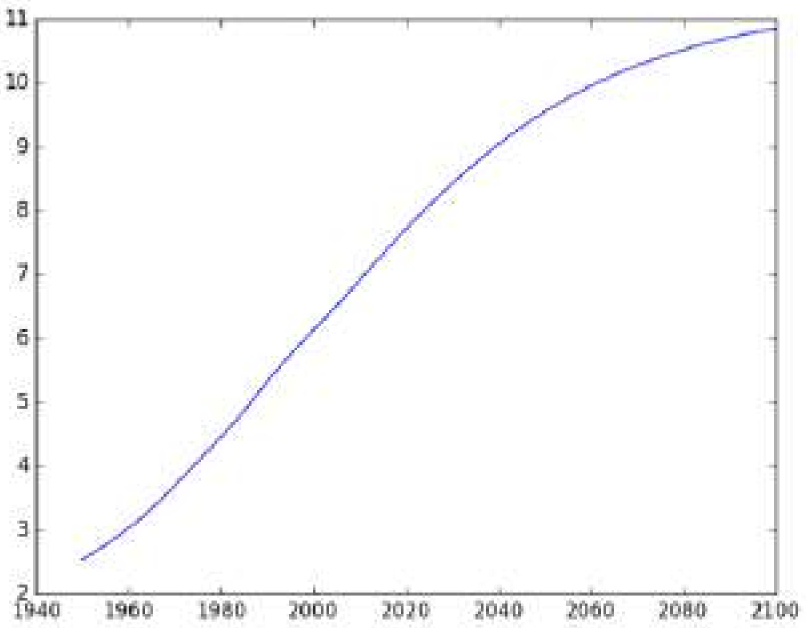
plt.plot(year, pop)

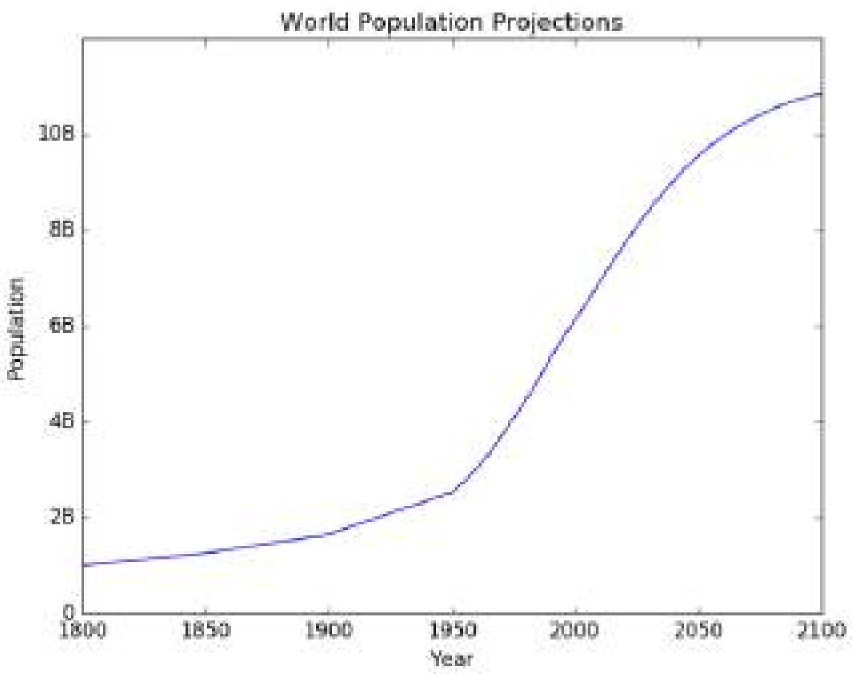
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10],
           ['0', '2B', '4B', '6B', '8B', '10B'])

plt.show()
```



Before vs. after





Let's practice!

INTERMEDIATE PYTHON

Dictionaries, Part 1

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List

```
pop = [30.55, 2.77, 39.21]
countries = ["afghanistan", "albania", "algeria"]
ind_alb = countries.index("albania")
ind_alb
```

```
1
```

```
pop[ind_alb]
```

```
2.77
```

- Not convenient
- Not intuitive

Dictionary

```
pop = [30.55, 2.77, 39.21]
countries = ["afghanistan", "albania", "algeria"]


...


        {                                                    }
```

Dictionary

```
pop = [30.55, 2.77, 39.21]
countries = ["afghanistan", "albania", "algeria"]


...


        {"afghanistan":30.55,                                    }
```

Dictionary

```
pop = [30.55, 2.77, 39.21]
countries = ["afghanistan", "albania", "algeria"]

...

world = {"afghanistan":30.55, "albania":2.77, "algeria":39.21}
world["albania"]
```

```
2.77
```

Let's practice!

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Dictionaries, Part 2

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Recap

```
world = {"afghanistan":30.55, "albania":2.77, "algeria":39.21}
world["albania"]
```

```
2.77
```

```
world = {"afghanistan":30.55, "albania":2.77,
         "algeria":39.21, "albania":2.81}
world
```

```
{'afghanistan': 30.55, 'albania': 2.81, 'algeria': 39.21}
```

Recap

- Keys have to be "immutable" objects

```
{0:"hello", True:"dear", "two":"world"}
```

```
{0: 'hello', True: 'dear', 'two': 'world'}
```

```
{["just", "to", "test"]: "value"}
```

```
TypeError: unhashable type: 'list'
```

Principality of Sealand

[1] Source: Wikipedia

Dictionary

```
world["sealand"] = 0.000027
world
```

```
{'afghanistan': 30.55, 'albania': 2.81,
        'algeria': 39.21, 'sealand': 2.7e-05}
```

```
"sealand" in world
```

```
True
```

Dictionary

```
world["sealand"] = 0.000028
world
```

```
{'afghanistan': 30.55, 'albania': 2.81,
        'algeria': 39.21, 'sealand': 2.8e-05}
```

```
del(world["sealand"])
world
```

```
{'afghanistan': 30.55, 'albania': 2.81, 'algeria': 39.21}
```

List vs. Dictionary

List vs. Dictionary

List vs. Dictionary

List	Dictionary
Select, update, and remove with `[]`	Select, update, and remove with `[]`

List vs. Dictionary

List	Dictionary
Select, update, and remove with `[]`	Select, update, and remove with `[]`

List vs. Dictionary

List	Dictionary
Select, update, and remove with `[]`	Select, update, and remove with `[]`
Indexed by range of numbers	

List vs. Dictionary

List	Dictionary
Select, update, and remove with `[]`	Select, update, and remove with `[]`
Indexed by range of numbers	Indexed by unique keys

List vs. Dictionary

List	Dictionary
Select, update, and remove with `[]`	Select, update, and remove with `[]`
Indexed by range of numbers	Indexed by unique keys
Collection of values — order matters, for selecting entire subsets	

List vs. Dictionary

List	Dictionary
Select, update, and remove with `[]`	Select, update, and remove with `[]`
Indexed by range of numbers	Indexed by unique keys
Collection of values — order matters, for selecting entire subsets	Lookup table with unique keys

Let's practice!

INTERMEDIATE PYTHON

Pandas, Part 1

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Tabular dataset examples

temperature	measured_at	location
76	2016-01-01 14:00:01	valve
86	2016-01-01 14:00:01	compressor
72	2016-01-01 15:00:01	valve
88	2016-01-01 15:00:01	compressor
68	2016-01-01 16:00:01	valve
78	2016-01-01 16:00:01	compressor

Tabular dataset examples

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row = observations
column = variable

Tabular dataset examples

temperature	measured_at	location
76	2016-01-01 14:00:01	valve
86	2016-01-01 14:00:01	compressor
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88	2016-01-01 15:00:01	compressor
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78	2016-01-01 16:00:01	compressor

row = observations
column = variable

country	capital	area	population
Brazil	Brasilia	8.516	200.4
Russia	Moscow	17.10	143.5
India	New Delhi	3.286	1252
China	Beijing	9.597	1357
South	Pretoria	1.221	52.98

Datasets in Python

- 2D NumPy array?
  - One data type

Datasets in Python

country	capital	area	population
Brazil	Brasilia	8.516	200.4
Russia	Moscow	17.10	143.5
India	New Delhi	3.286	1252
China	Beijing	9.597	1357
South	Pretoria	1.221	52.98
		float	float

Datasets in Python

country	capital	area	population
Brazil	Brasilia	8.516	200.4
Russia	Moscow	17.10	143.5
India	New Delhi	3.286	1252
China	Beijing	9.597	1357
South	Pretoria	1.221	52.98
str	str	float	float

- pandas!
  - High level data manipulation tool
  - Wes McKinney
  - Built on NumPy
  - DataFrame

DataFrame

```
brics
```

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

DataFrame from Dictionary

```
dict = {
    "country":["Brazil", "Russia", "India", "China", "South Africa"],
    "capital":["Brasilia", "Moscow", "New Delhi", "Beijing", "Pretoria"],
       "area":[8.516, 17.10, 3.286, 9.597, 1.221]
 "population":[200.4, 143.5, 1252, 1357, 52.98] }
```

- keys (column labels)
- values (data, column by column)

```
import pandas as pd
brics = pd.DataFrame(dict)
```

DataFrame from Dictionary (2)

```
brics
```

```
    area    capital       country  population
0   8.516   Brasilia       Brazil      200.40
1  17.100     Moscow       Russia      143.50
2   3.286  New Delhi        India     1252.00
3   9.597    Beijing        China     1357.00
4   1.221   Pretoria  South Africa      52.98
```

```
brics.index = ["BR", "RU", "IN", "CH", "SA"]
brics
```

```
     area    capital       country  population
BR   8.516   Brasilia       Brazil      200.40
RU  17.100     Moscow       Russia      143.50
IN   3.286  New Delhi        India     1252.00
CH   9.597    Beijing        China     1357.00
SA   1.221   Pretoria  South Africa      52.98
```

DataFrame from CSV file

`brics.csv`

```
,country,capital,area,population
BR,Brazil,Brasilia,8.516,200.4
RU,Russia,Moscow,17.10,143.5
IN,India,New Delhi,3.286,1252
CH,China,Beijing,9.597,1357
SA,South Africa,Pretoria,1.221,52.98
```

- CSV = comma-separated values

DataFrame from CSV file

- `brics.csv`

```
,country,capital,area,population
BR,Brazil,Brasilia,8.516,200.4
RU,Russia,Moscow,17.10,143.5
IN,India,New Delhi,3.286,1252
CH,China,Beijing,9.597,1357
SA,South Africa,Pretoria,1.221,52.98
```

```
brics = pd.read_csv("path/to/brics.csv")
brics
```

```
  Unnamed: 0       country    capital    area  population
0         BR        Brazil   Brasilia   8.516      200.40
1         RU        Russia     Moscow  17.100      143.50
2         IN         India  New Delhi   3.286     1252.00
3         CH         China    Beijing   9.597     1357.00
4         SA  South Africa   Pretoria   1.221       52.98
```

DataFrame from CSV file

```
brics = pd.read_csv("path/to/brics.csv", index_col = 0)
brics
```

```
         country  population      area    capital
BR        Brazil         200   8515767   Brasilia
RU        Russia         144  17098242     Moscow
IN         India        1252   3287590  New Delhi
CH         China        1357   9596961    Beijing
SA  South Africa          55   1221037   Pretoria
```

Let's practice!

INTERMEDIATE PYTHON

Pandas, Part 2

INTERMEDIATE PYTHON

Hugo Bowne-Anderson
Data Scientist at DataCamp

brics

```
import pandas as pd
brics = pd.read_csv("path/to/brics.csv", index_col = 0)
brics
```

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

Index and select data

- Square brackets
- Advanced methods
  - loc
  - iloc

Column Access [ ]

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics["country"]
```

```
BR          Brazil
RU          Russia
IN           India
CH           China
SA    South Africa
Name: country, dtype: object
```

Column Access [ ]

```
         country    capital    area   population
BR        Brazil   Brasilia   8.516       200.40
RU        Russia     Moscow  17.100       143.50
IN         India  New Delhi   3.286      1252.00
CH         China    Beijing   9.597      1357.00
SA  South Africa   Pretoria   1.221        52.98
```

```
type(brics["country"])
```

```
pandas.core.series.Series
```

- 1D labelled array

Column Access [ ]

```
     country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics[["country"]]
```

```
         country
BR        Brazil
RU        Russia
IN         India
CH         China
SA  South Africa
```

Column Access [ ]

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```python
type(brics[["country"]])
```

```
pandas.core.frame.DataFrame
```

Column Access [ ]

```
       country    capital    area  population
BR      Brazil   Brasilia   8.516      200.40
RU      Russia     Moscow  17.100      143.50
IN       India  New Delhi   3.286     1252.00
CH       China    Beijing   9.597     1357.00
SA  South Africa  Pretoria  1.221       52.98
```

```
brics[["country", "capital"]]
```

```
       country    capital
BR      Brazil   Brasilia
RU      Russia     Moscow
IN       India  New Delhi
CH       China    Beijing
SA  South Africa  Pretoria
```

Row Access [ ]

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics[1:4]
```

```
   country    capital    area  population
RU  Russia     Moscow  17.100       143.5
IN   India  New Delhi   3.286      1252.0
CH   China    Beijing   9.597      1357.0
```

Row Access [ ]

```
        country    capital    area  population
BR       Brazil   Brasilia   8.516      200.40    * 0 *
RU       Russia     Moscow  17.100      143.50    * 1 *
IN        India  New Delhi   3.286     1252.00    * 2 *
CH        China    Beijing   9.597     1357.00    * 3 *
SA  South Africa  Pretoria   1.221       52.98    * 4 *
```

```
brics[1:4]
```

```
   country    capital    area  population
RU  Russia     Moscow  17.100       143.5
IN   India  New Delhi   3.286      1252.0
CH   China    Beijing   9.597      1357.0
```

Discussion [ ]

- Square brackets: limited functionality
- Ideally
  - 2D NumPy arrays
  - `my_array[rows, columns]`
- pandas
  - `loc` (label-based)
  - `iloc` (integer position-based)

Row Access loc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.loc["RU"]
```

```
country       Russia
capital       Moscow
area            17.1
population     143.5
Name: RU, dtype: object
```

- Row as pandas Series

Row Access loc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.loc[["RU"]]
```

```
   country capital  area  population
RU  Russia  Moscow  17.1       143.5
```

- DataFrame

Row Access loc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.loc[["RU", "IN", "CH"]]
```

```
   country    capital    area  population
RU  Russia     Moscow  17.100       143.5
IN   India  New Delhi   3.286      1252.0
CH   China    Beijing   9.597      1357.0
```

Row & Column loc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.loc[["RU", "IN", "CH"], ["country", "capital"]]
```

```
   country    capital
RU  Russia     Moscow
IN   India  New Delhi
CH   China    Beijing
```

Row & Column loc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.loc[:, ["country", "capital"]]
```

```
         country    capital
BR        Brazil   Brasilia
RU        Russia     Moscow
IN         India  New Delhi
CH         China    Beijing
SA  South Africa   Pretoria
```

Recap

- Square brackets
  - Column access `brics[["country", "capital"]]`
  - Row access: only through slicing `brics[1:4]`
- loc (label-based)
  - Row access `brics.loc[["RU", "IN", "CH"]]`
  - Column access `brics.loc[:, ["country", "capital"]]`
  - Row & Column access

```
brics.loc[
["RU", "IN", "CH"],
["country", "capital"]
]
```

Row Access iloc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.loc[["RU"]]
```

```
   country capital  area  population
RU  Russia  Moscow  17.1       143.5
```

```
brics.iloc[[1]]
```

```
   country capital  area  population
RU  Russia  Moscow  17.1       143.5
```

Row Access iloc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.loc[["RU", "IN", "CH"]]
```

```
   country    capital    area  population
RU  Russia     Moscow  17.100       143.5
IN   India  New Delhi   3.286      1252.0
CH   China    Beijing   9.597      1357.0
```

Row Access iloc

```
          country    capital    area  population
BR         Brazil   Brasilia   8.516      200.40
RU         Russia     Moscow  17.100      143.50
IN          India  New Delhi   3.286     1252.00
CH          China    Beijing   9.597     1357.00
SA   South Africa   Pretoria   1.221       52.98
```

```
brics.iloc[[1,2,3]]
```

```
    country    capital    area  population
RU   Russia     Moscow  17.100       143.5
IN    India  New Delhi   3.286      1252.0
CH    China    Beijing   9.597      1357.0
```

Row & Column iloc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.loc[["RU", "IN", "CH"], ["country", "capital"]]
```

```
   country    capital
RU  Russia     Moscow
IN   India  New Delhi
CH   China    Beijing
```

Row & Column iloc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics.iloc[[1,2,3], [0, 1]]
```

```
   country    capital
RU  Russia     Moscow
IN   India  New Delhi
CH   China    Beijing
```

Row & Column iloc

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```python
brics.loc[:, ["country", "capital"]]
```

```
         country    capital
BR        Brazil   Brasilia
RU        Russia     Moscow
IN         India  New Delhi
CH         China    Beijing
SA  South Africa   Pretoria
```

Row & Column iloc

```
      country    capital    area  population
BR          Brazil   Brasilia   8.516      200.40
RU          Russia     Moscow  17.100      143.50
IN           India  New Delhi   3.286     1252.00
CH           China    Beijing   9.597     1357.00
SA    South Africa   Pretoria   1.221       52.98
```

```
brics.iloc[:, [0,1]]
```

```
         country    capital
BR        Brazil   Brasilia
RU        Russia     Moscow
IN         India  New Delhi
CH         China    Beijing
SA  South Africa   Pretoria
```

Let's practice!

INTERMEDIATE PYTHON

Comparison Operators

INTERMEDIATE PYTHON

Hugo Bowne-Anderson
Data Scientist at DataCamp

NumPy recap

```python
# Code from Intro to Python for Data Science, Chapter 4
import numpy as np
np_height = np.array([1.73, 1.68, 1.71, 1.89, 1.79])
np_weight = np.array([65.4, 59.2, 63.6, 88.4, 68.7])
bmi = np_weight / np_height ** 2
bmi
```

```
array([ 21.852,  20.975,  21.75 ,  24.747,  21.441])
```

```python
bmi > 23
```

```
array([False, False, False,  True, False], dtype=bool)
```

```python
bmi[bmi > 23]
```

```
array([ 24.747])
```

- Comparison operators: how Python values relate

Numeric comparisons

```
2 < 3
```

```
True
```

```
2 == 3
```

```
False
```

```
2 <= 3
```

```
True
```

```
3 <= 3
```

```
True
```

```
x = 2
y = 3
x < y
```

```
True
```

Other comparisons

```
"carl" < "chris"
```

```
True
```

```
3 < "chris"
```

```
TypeError: unorderable types: int() < str()
```

```
3 < 4.1
```

```
True
```

Other comparisons

```
bmi
```

```
array([21.852, 20.975, 21.75 , 24.747, 21.441])
```

```
bmi > 23
```

```
array([False, False, False, True, False], dtype=bool)
```

Comparators

Comparator	Meaning
`<`	Strictly less than
`<=`	Less than or equal
`>`	Strictly greater than
`>=`	Greater than or equal
`==`	Equal
`!=`	Not equal

Let's practice!

INTERMEDIATE PYTHON

Boolean Operators

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Boolean Operators

- `and`
- `or`
- `not`

and

```
True and True
```

```
True
```

```
x = 12
x > 5 and x < 15
# True      True
```

```
True
```

```
False and True
```

```
False
```

```
True and False
```

```
False
```

```
False and False
```

```
False
```

or

```python
True or True
```

```
True
```

```python
False or True
```

```
True
```

```python
True or False
```

```
True
```

```python
False or False
```

```
False
```

```python
y = 5
y < 7 or y > 13
```

```
True
```

not

```
not True
```

```
False
```

```
not False
```

```
True
```

NumPy

```
bmi     # calculation of bmi left out
```

```
array([21.852, 20.975, 21.75 , 24.747, 21.441])
```

```
bmi > 21
```

```
array([True, False, True, True, True], dtype=bool)
```

```
bmi < 22
```

```
array([True, True, True, False, True], dtype=bool)
```

```
bmi > 21 and bmi < 22
```

```
ValueError: The truth value of an array with more than one element is
ambiguous. Use a.any() or a.all()
```

NumPy

- `logical_and()`
- `logical_or()`
- `logical_not()`

```
np.logical_and(bmi > 21, bmi < 22)
```

```
array([True, False, True, False, True], dtype=bool)
```

```
bmi[np.logical_and(bmi > 21, bmi < 22)]
```

```
array([21.852, 21.75, 21.441])
```

Let's practice!

INTERMEDIATE PYTHON

if, elif, else

INTERMEDIATE PYTHON

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Overview

- Comparison Operators
  - `<` , `>` , `>=` , `<=` , `==` , `!=`
- Boolean Operators
  - `and` , `or` , `not`
- Conditional Statements
  - `if` , `else` , `elif`

if

```
if condition :
    expression
```

`control.py`

```
z = 4
if z % 2 == 0 :     # True
    print("z is even")
```

```
z is even
```

if

```
if condition :
    expression
```

- `expression` not part of if

`control.py`

```
z = 4
if z % 2 == 0 :    # True
    print("z is even")
```

```
z is even
```

if

```
if condition :
    expression
```

`control.py`

```
z = 4
if z % 2 == 0 :
    print("checking "  + str(z))
    print("z is even")
```

```
checking 4
z is even
```

if

```
if condition :
    expression
```

`control.py`

```
z = 5
if z % 2 == 0 :    # False
    print("checking "  + str(z))
    print("z is even")
```

else

```
if condition :
    expression
else :
    expression
```

`control.py`

```
z = 5
if z % 2 == 0 :    # False
    print("z is even")
else :
    print("z is odd")
```

```
z is odd
```

elif

```
if condition :
    expression
elif condition :
    expression
else :
    expression
```

`control.py`

```
z = 3
if z % 2 == 0 :
    print("z is divisible by 2")    # False
elif z % 3 == 0 :
    print("z is divisible by 3")    # True
else :
    print("z is neither divisible by 2 nor by 3")
```

```
z is divisible by 3
```

elif

```
if condition :
    expression
elif condition :
    expression
else :
    expression
```

`control.py`

```
z = 6
if z % 2 == 0 :
    print("z is divisible by 2")    # True
elif z % 3 == 0 :
    print("z is divisible by 3")    # Never reached
else :
    print("z is neither divisible by 2 nor by 3")
```

```
z is divisible by 2
```

Let's practice!

INTERMEDIATE PYTHON

Filtering pandas DataFrames

INTERMEDIATE PYTHON

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brics

```python
import pandas as pd
brics = pd.read_csv("path/to/brics.csv", index_col = 0)
brics
```

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

Goal

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

- Select countries with area over 8 million km2
- 3 steps
    - Select the area column
    - Do comparison on area column
    - Use result to select countries

Step 1: Get column

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```
brics["area"]
```

```
BR     8.516
RU    17.100
IN     3.286
CH     9.597
SA     1.221
Name: area, dtype: float64    # - Need Pandas Series
```

- Alternatives:

```
brics.loc[:,"area"]
brics.iloc[:,2]
```

Step 2: Compare

```
brics["area"]
```

```
BR     8.516
RU    17.100
IN     3.286
CH     9.597
SA     1.221
Name: area, dtype: float64
```

```
brics["area"] > 8
```

```
BR     True
RU     True
IN    False
CH     True
SA    False
Name: area, dtype: bool
```

```
is_huge = brics["area"] > 8
```

Step 3: Subset DF

```
is_huge
```

```
BR     True
RU     True
IN    False
CH     True
SA    False
Name: area, dtype: bool
```

```
brics[is_huge]
```

```
   country   capital    area  population
BR  Brazil  Brasilia   8.516       200.4
RU  Russia    Moscow  17.100       143.5
CH   China   Beijing   9.597      1357.0
```

Summary

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.988
```

```
is_huge = brics["area"] > 8
brics[is_huge]
```

```
   country   capital    area  population
BR  Brazil  Brasilia   8.516       200.4
RU  Russia    Moscow  17.100       143.5
CH   China   Beijing   9.597      1357.0
```

```
brics[brics["area"] > 8]
```

```
   country   capital    area  population
BR  Brazil  Brasilia   8.516       200.4
RU  Russia    Moscow  17.100       143.5
CH   China   Beijing   9.597      1357.0
```

Boolean operators

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

```python
import numpy as np
np.logical_and(brics["area"] > 8, brics["area"] < 10)
```

```
BR      True
RU     False
IN     False
CH      True
SA     False
Name: area, dtype: bool
```

```python
brics[np.logical_and(brics["area"] > 8, brics["area"] < 10)]
```

```
   country   capital   area  population
BR  Brazil  Brasilia  8.516       200.4
CH   China   Beijing  9.597      1357.0
```

Let's practice!

INTERMEDIATE PYTHON

while loop

INTERMEDIATE PYTHON

Hugo Bowne-Anderson
Data Scientist at DataCamp

if-elif-else

`control.py`

- Goes through construct only once!

```
z = 6
if z % 2 == 0 :  # True
    print("z is divisible by 2")  # Executed
elif z % 3 == 0 :
    print("z is divisible by 3")
else :
    print("z is neither divisible by 2 nor by 3")

...   # Moving on
```

- While loop = repeated if statement

While

```
while condition :
    expression
```

- Numerically calculating model
- "repeating action until condition is met"
- Example
  - Error starts at 50
  - Divide error by 4 on every run
  - Continue until error no longer > 1

While

```
while condition :
    expression
```

`while_loop.py`

```python
error = 50.0

while error > 1:
    error = error / 4
    print(error)
```

- Error starts at 50
- Divide error by 4 on every run
- Continue until error no longer > 1

While

```
while condition :
    expression
```

`while_loop.py`

```
error = 50.0
#      50
while error > 1:     # True
      error = error / 4
      print(error)
```

```
12.5
```

While

```
while condition :
    expression
```

`while_loop.py`

```
error = 50.0
#      12.5
while error > 1:    # True
      error = error / 4
      print(error)
```

```
12.5
3.125
```

While

```
while condition :
    expression
```

`while_loop.py`

```
error = 50.0
#      3.125
while error > 1:    # True
      error = error / 4
      print(error)
```

```
12.5
3.125
0.78125
```

While

```
while condition :
    expression
```

`while_loop.py`

```
error = 50.0
#      0.78125
while error > 1:    # False
    error = error / 4
    print(error)
```

```
12.5
3.125
0.78125
```

While

```
while condition :
    expression
```

`while_loop.py`

```
error = 50.0
while error > 1 :    # always True
    # error = error / 4
    print(error)
```

```
50
50
50
50
50
50
50
...
```

- DataCamp: session disconnected
- Local system: Control + C

Let's practice!

INTERMEDIATE PYTHON

for loop

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Data Scientist at DataCamp

for loop

```
for var in seq :
    expression
```

- "for each var in seq, execute expression"

fam

`family.py`

```python
fam = [1.73, 1.68, 1.71, 1.89]
print(fam)
```

```
[1.73, 1.68, 1.71, 1.89]
```

fam

`family.py`

```
fam = [1.73, 1.68, 1.71, 1.89]
print(fam[0])
print(fam[1])
print(fam[2])
print(fam[3])
```

```
1.73
1.68
1.71
1.89
```

for loop

```python
for var in seq :
    expression
```

`family.py`

```python
fam = [1.73, 1.68, 1.71, 1.89]
for height in fam :
    print(height)
```

for loop

```
for var in seq :
    expression
```

`family.py`

```
fam = [1.73, 1.68, 1.71, 1.89]
for height in fam :
    print(height)
    # first iteration
    # height = 1.73
```

```
1.73
```

for loop

```
for var in seq :
    expression
```

`family.py`

```
fam = [1.73, 1.68, 1.71, 1.89]
for height in fam :
    print(height)
    # second iteration
    # height = 1.68
```

```
1.73
1.68
```

for loop

```
for var in seq :
    expression
```

`family.py`

```
fam = [1.73, 1.68, 1.71, 1.89]
for height in fam :
    print(height)
```

```
1.73
1.68
1.71
1.89
```

- No access to indexes

for loop

```
for var in seq :
    expression
```

`family.py`

```
fam = [1.73, 1.68, 1.71, 1.89]
```

- ???

```
index 0: 1.73
index 1: 1.68
index 2: 1.71
index 3: 1.89
```

enumerate

```
for var in seq :
    expression
```

`family.py`

```
fam = [1.73, 1.68, 1.71, 1.89]
for index, height in enumerate(fam) :
    print("index " + str(index) + ": " + str(height))
```

```
index 0: 1.73
index 1: 1.68
index 2: 1.71
index 3: 1.89
```

Loop over string

```
for var in seq :
    expression
```

`strloop.py`

```
for c in "family" :
    print(c.capitalize())
```

```
F
A
M
I
L
Y
```

Let's practice!

INTERMEDIATE PYTHON

Loop Data Structures Part 1

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Dictionary

```
for var in seq :
    expression
```

`dictloop.py`

```
world = { "afghanistan":30.55,
          "albania":2.77,
          "algeria":39.21 }
for key, value in world :
    print(key + " -- " + str(value))
```

```
ValueError: too many values to
            unpack (expected 2)
```

Dictionary

```
for var in seq :
    expression
```

`dictloop.py`

```
world = { "afghanistan":30.55,
          "albania":2.77,
          "algeria":39.21 }
for key, value in world.items() :
    print(key + " -- " + str(value))
```

```
algeria -- 39.21
afghanistan -- 30.55
albania -- 2.77
```

Dictionary

```
for var in seq :
    expression
```

`dictloop.py`

```
world = { "afghanistan":30.55,
          "albania":2.77,
          "algeria":39.21 }
for k, v in world.items() :
    print(k + " -- " + str(v))
```

```
algeria -- 39.21
afghanistan -- 30.55
albania -- 2.77
```

NumPy Arrays

```
for var in seq :
    expression
```

`nploop.py`

```
import numpy as np
np_height = np.array([1.73, 1.68, 1.71, 1.89, 1.79])
np_weight = np.array([65.4, 59.2, 63.6, 88.4, 68.7])
bmi = np_weight / np_height ** 2
for val in bmi :
    print(val)
```

```
21.852
20.975
21.750
24.747
21.441
```

2D NumPy Arrays

`nploop.py`

```python
import numpy as np
np_height = np.array([1.73, 1.68, 1.71, 1.89, 1.79])
np_weight = np.array([65.4, 59.2, 63.6, 88.4, 68.7])
meas = np.array([np_height, np_weight])
for val in meas :
    print(val)
```

```
[ 1.73  1.68  1.71  1.89  1.79]
[ 65.4  59.2  63.6  88.4  68.7]
```

2D NumPy Arrays

`nploop.py`

```python
import numpy as np
np_height = np.array([1.73, 1.68, 1.71, 1.89, 1.79])
np_weight = np.array([65.4, 59.2, 63.6, 88.4, 68.7])
meas = np.array([np_height, np_weight])
for val in np.nditer(meas) :
    print(val)
```

```
1.73
1.68
1.71
1.89
1.79
65.4
...
```

Recap

- Dictionary
  - for `key, val in my_dict.items() :`
- NumPy array
  - `for val in np.nditer(my_array) :`

Let's practice!

INTERMEDIATE PYTHON

Loop Data Structures Part 2

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Data Scientist at DataCamp

brics

```
         country    capital    area  population
BR        Brazil   Brasilia   8.516      200.40
RU        Russia     Moscow  17.100      143.50
IN         India  New Delhi   3.286     1252.00
CH         China    Beijing   9.597     1357.00
SA  South Africa   Pretoria   1.221       52.98
```

`dfloop.py`

```python
import pandas as pd
brics = pd.read_csv("brics.csv", index_col = 0)
```

for, first try

`dfloop.py`

```
import pandas as pd
brics = pd.read_csv("brics.csv", index_col = 0)
for val in brics :
    print(val)
```

```
country
capital
area
population
```

iterrows

dfloop.py

```
import pandas as pd
brics = pd.read_csv("brics.csv", index_col = 0)
for lab, row in brics.iterrows():
    print(lab)
    print(row)
```

```
BR
country           Brazil
capital         Brasilia
area               8.516
population         200.4
Name: BR, dtype: object
...
RU
country         Russia
capital         Moscow
area              17.1
population       143.5
Name: RU, dtype: object
IN ...
```

Selective print

`dfloop.py`

```python
import pandas as pd
brics = pd.read_csv("brics.csv", index_col = 0)
for lab, row in brics.iterrows():
    print(lab + ": " + row["capital"])
```

```
BR: Brasilia
RU: Moscow
IN: New Delhi
CH: Beijing
SA: Pretoria
```

Add column

`dfloop.py`

```python
import pandas as pd
brics = pd.read_csv("brics.csv", index_col = 0)
for lab, row in brics.iterrows() :
    # - Creating Series on every iteration
    brics.loc[lab, "name_length"] = len(row["country"])
print(brics)
```

```
         country    capital    area  population  name_length
BR        Brazil   Brasilia   8.516      200.40            6
RU        Russia     Moscow  17.100      143.50            6
IN         India  New Delhi   3.286     1252.00            5
CH         China    Beijing   9.597     1357.00            5
SA  South Africa   Pretoria   1.221       52.98           12
```

apply

`dfloop.py`

```python
import pandas as pd
brics = pd.read_csv("brics.csv", index_col = 0)
brics["name_length"] = brics["country"].apply(len)
print(brics)
```

```
         country    capital    area  population  name_length
BR        Brazil   Brasilia   8.516      200.40            6
RU        Russia     Moscow  17.100      143.50            6
IN         India  New Delhi   3.286     1252.00            5
CH         China    Beijing   9.597     1357.00            5
SA  South Africa   Pretoria   1.221       52.98           12
```

Let's practice!

INTERMEDIATE PYTHON

Random Numbers

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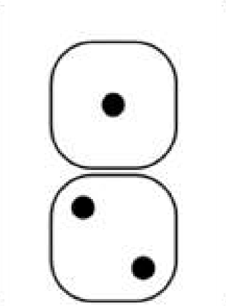
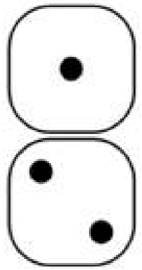
100 x

100 x

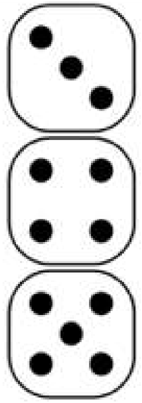


-1

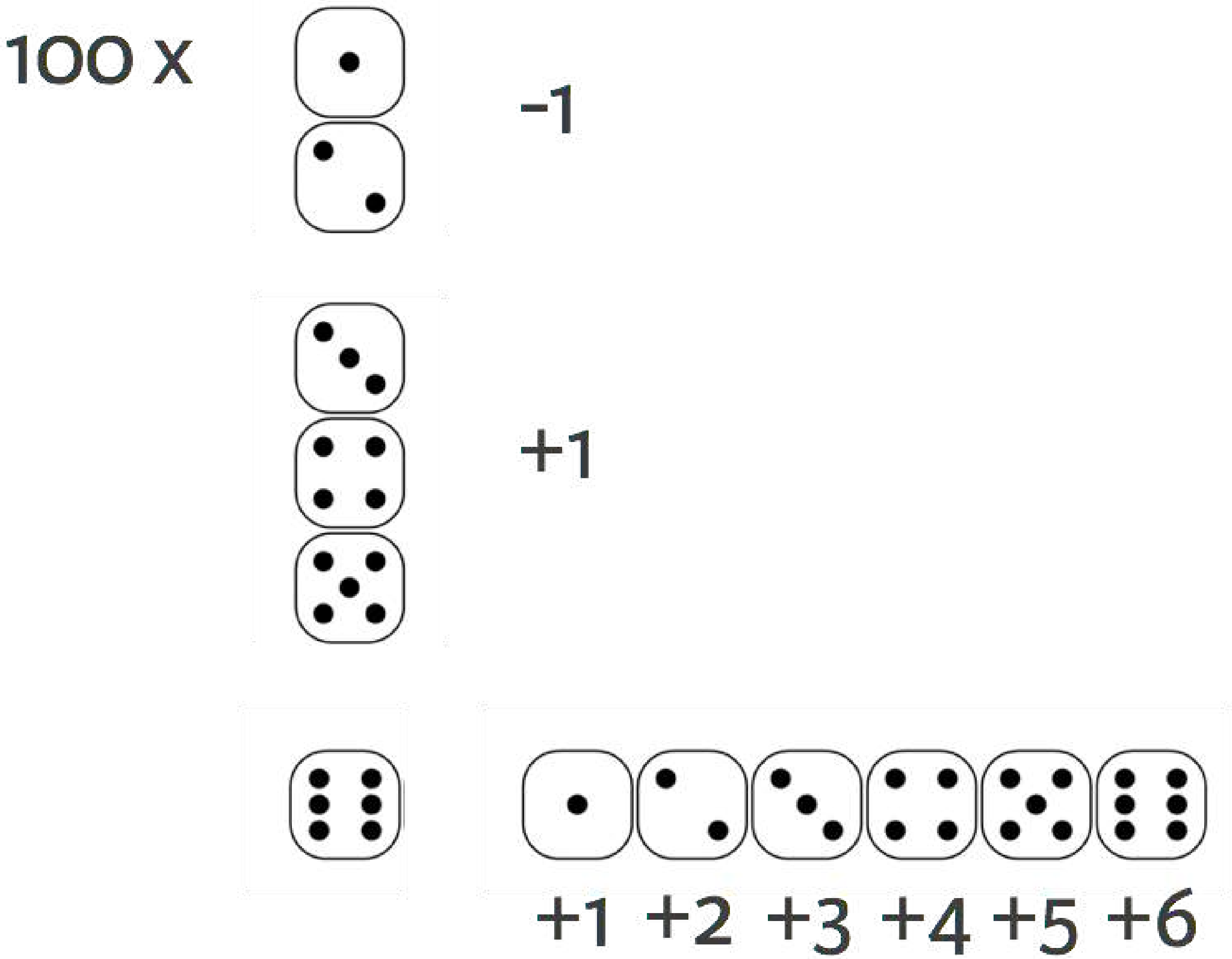
100 x

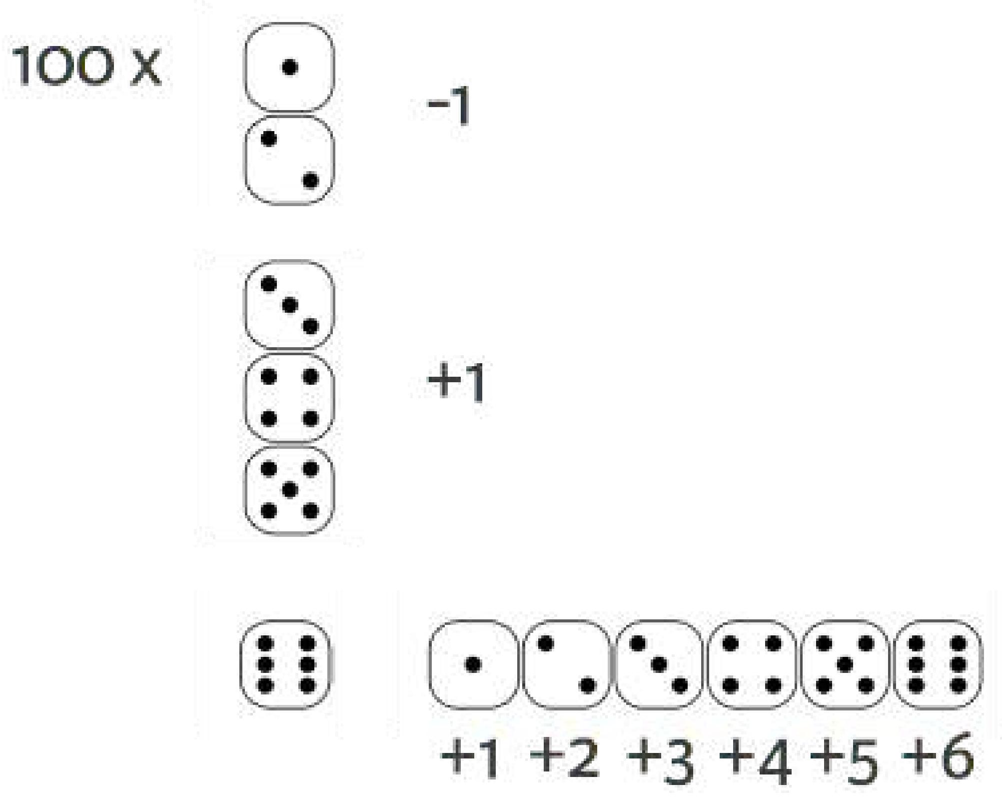


-1



+1


100 x
-1
+1
+1 +2 +3 +4 +5 +6



- Can't go below step 0
- 0.1 % chance of falling down the stairs
- Bet: you'll reach step 60

How to solve?

- Analytical
- Simulate the process
  - Hacker statistics!

Random generators

```
import numpy as np
np.random.rand()       # Pseudo-random numbers
```

```
0.9535543896720104     # Mathematical formula
```

```
np.random.seed(123)     # Starting from a seed
np.random.rand()
```

```
0.6964691855978616
```

```
np.random.rand()
```

```
0.28613933495037946
```

Random generators

```
np.random.seed(123)
np.random.rand()
```

```
0.696469185597861    # Same seed: same random numbers!
```

```
np.random.rand()     # Ensures "reproducibility"
```

```
0.28613933495037946
```

Coin toss

`game.py`

```python
import numpy as np
np.random.seed(123)
coin = np.random.randint(0,2)  # Randomly generate 0 or 1
print(coin)
```

```
0
```

Coin toss

`game.py`

```python
import numpy as np
np.random.seed(123)
coin = np.random.randint(0,2)  # Randomly generate 0 or 1
print(coin)
if coin == 0:
    print("heads")
else:
    print("tails")
```

```
0
heads
```

Let's practice!

INTERMEDIATE PYTHON

Random Walk

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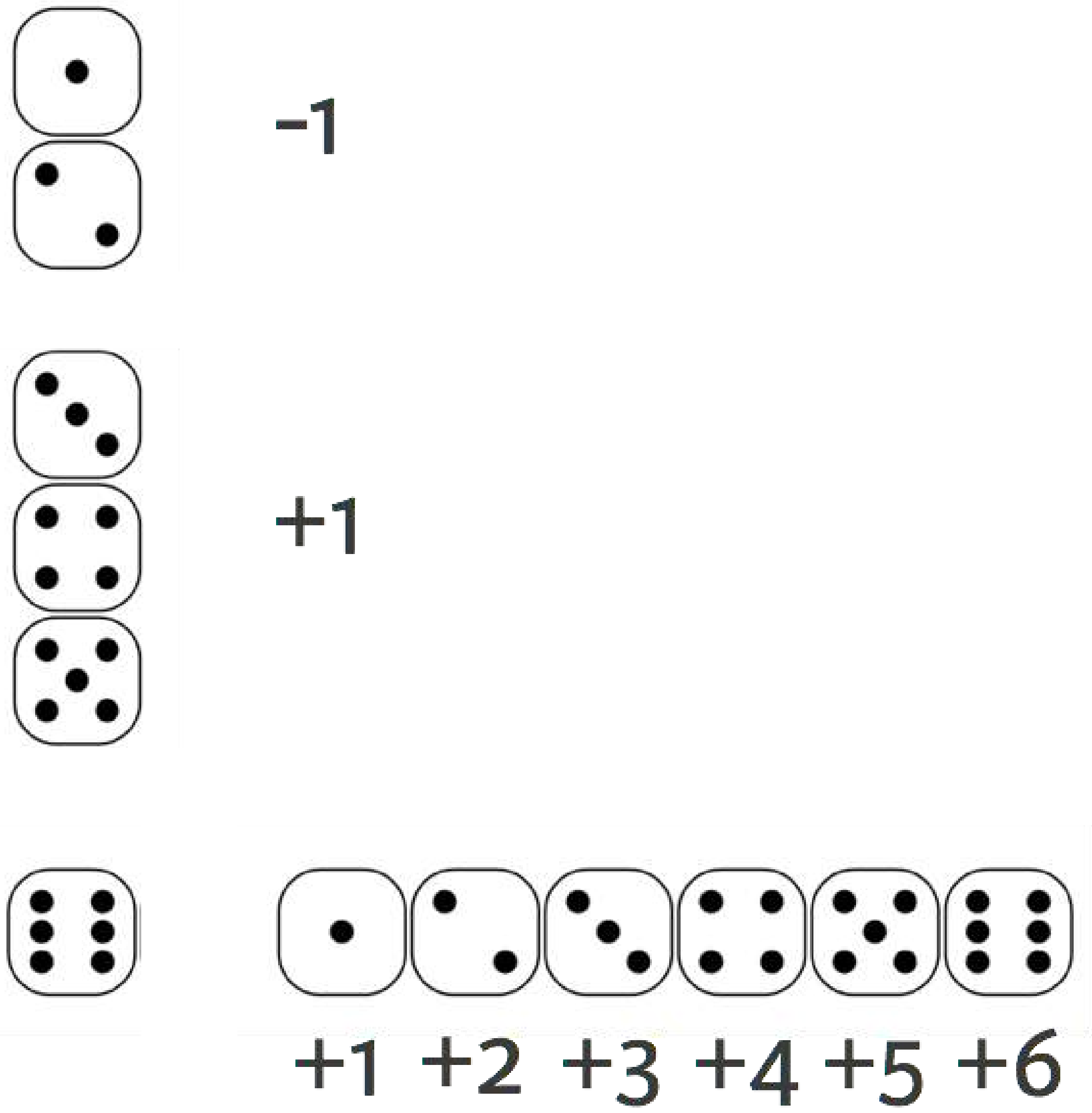
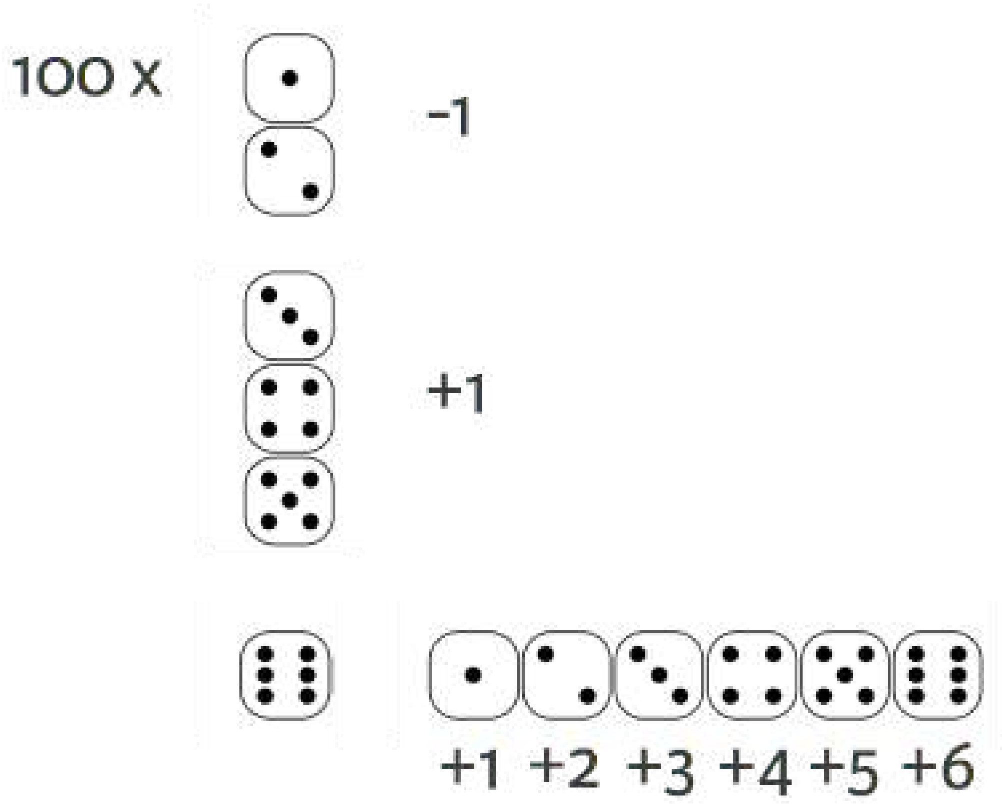
Random Step



Random Walk



Known in Science

- Path of molecules
- Gambler's financial status

Heads or Tails

`headtails.py`

```python
import numpy as np
np.random.seed(123)
outcomes = []
for x in range(10) :
    coin = np.random.randint(0, 2)
    if coin == 0 :
        outcomes.append("heads")
    else :
        outcomes.append("tails")
print(outcomes)
```

```
['heads', 'tails', 'heads', 'heads', 'heads',
 'heads', 'heads', 'tails', 'tails', 'heads']
```

Heads or Tails: Random Walk

`headtailsrw.py`

```
import numpy as np
np.random.seed(123)
tails = [0]
for x in range(10) :
    coin = np.random.randint(0, 2)
    tails.append(tails[x] + coin)
print(tails)
```

```
[0, 0, 1, 1, 1, 1, 1, 1, 2, 3, 3]
```

Step to Walk

```
outcomes
```

```
['heads', 'tails', 'heads', 'heads', 'heads',
 'heads', 'heads', 'tails', 'tails', 'heads']
```

```
tails
```

```
[0, 0, 1, 1, 1, 1, 1, 1, 2, 3, 3]
```

Let's practice!

INTERMEDIATE PYTHON

Distribution

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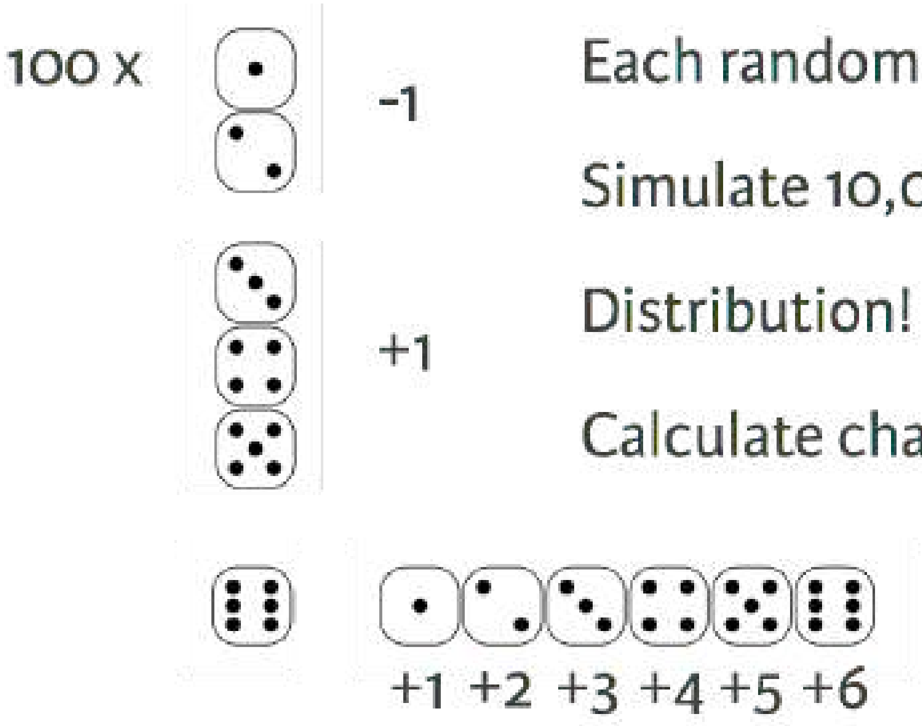
Distribution



Each random walk has an end point

Simulate 10,000 times: 10,000 end points

Distribution!

Calculate chances!

Random Walk

`headtailsrw.py`

```
import numpy as np
np.random.seed(123)
tails = [0]
for x in range(10) :
    coin = np.random.randint(0, 2)
    tails.append(tails[x] + coin)
```

100 runs

`distribution.py`

```python
import numpy as np
np.random.seed(123)
final_tails = []
for x in range(100) :
    tails = [0]
    for x in range(10) :
        coin = np.random.randint(0, 2)
        tails.append(tails[x] + coin)
    final_tails.append(tails[-1])
print(final_tails)
```

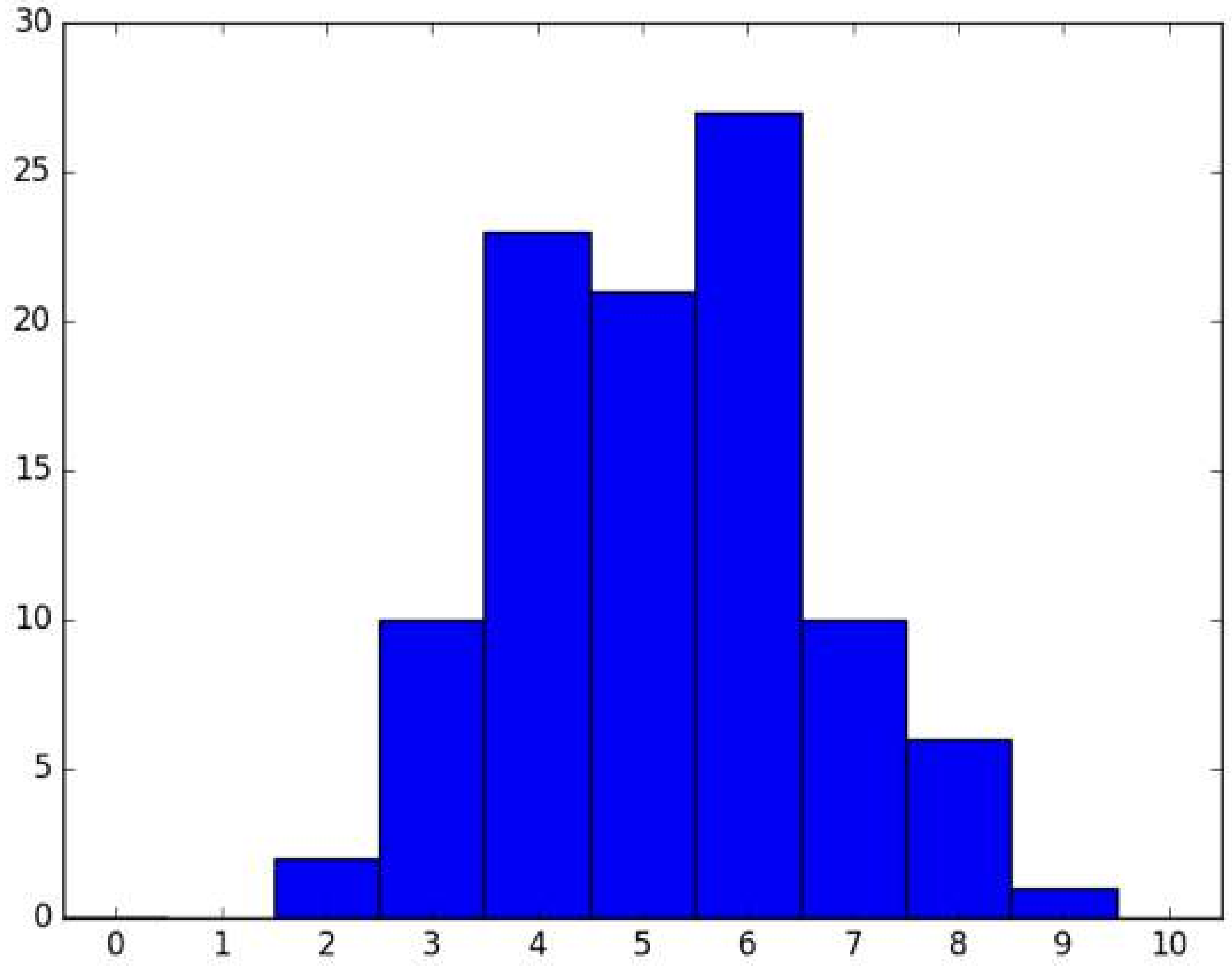
```
[3, 6, 4, 5, 4, 5, 3, 5, 4, 6, 6, 8, 6, 4, 7, 5, 7, 4, 3, 3, ..., 4]
```

Histogram, 100 runs

`distribution.py`

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(123)
final_tails = []
for x in range(100) :
    tails = [0]
    for x in range(10) :
        coin = np.random.randint(0, 2)
        tails.append(tails[x] + coin)
    final_tails.append(tails[-1])
plt.hist(final_tails, bins = 10)
plt.show()
```

Histogram, 100 runs

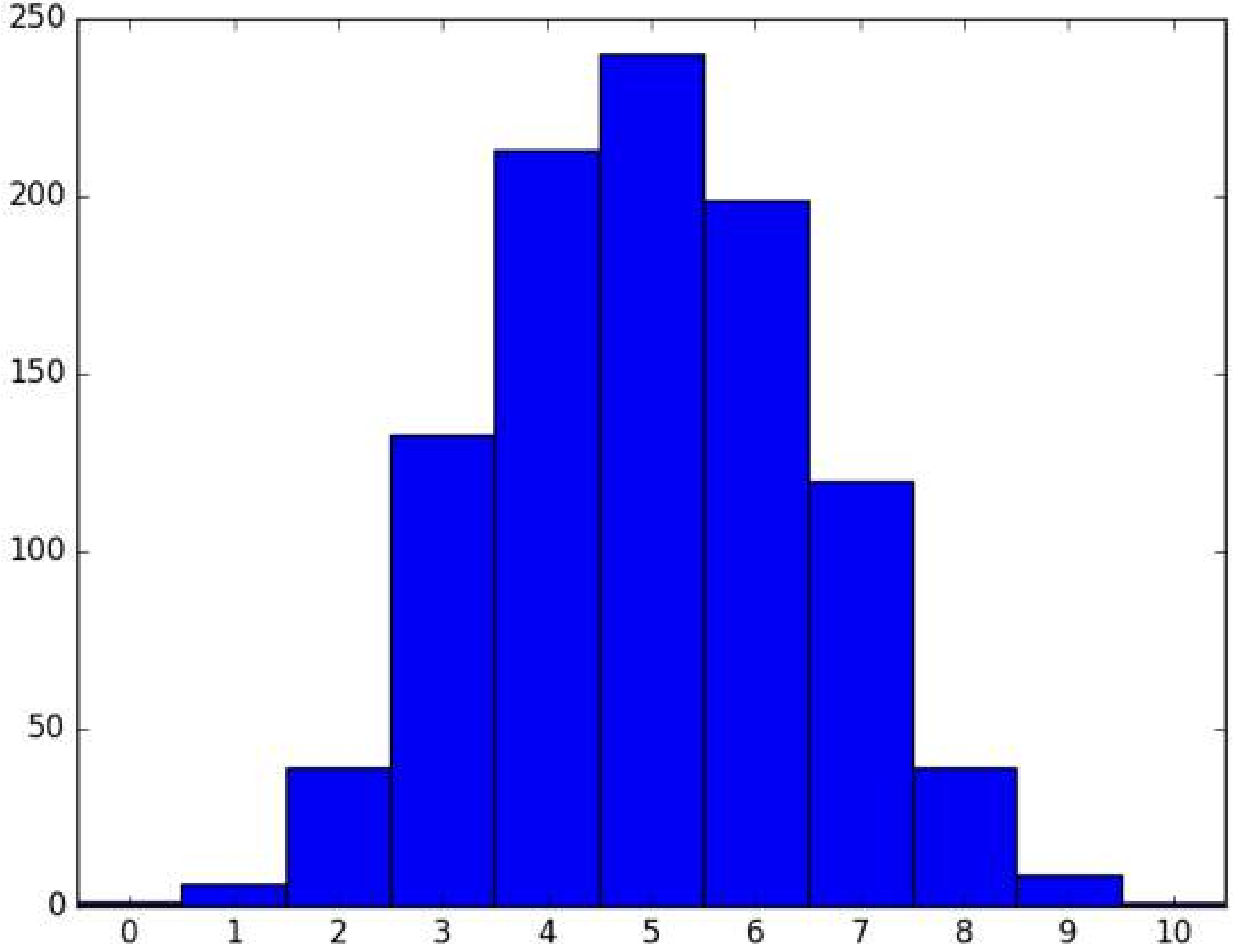


Histogram, 1,000 runs

`distribution.py`

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(123)
final_tails = []
for x in range(1000) : # <--
    tails = [0]
    for x in range(10) :
        coin = np.random.randint(0, 2)
        tails.append(tails[x] + coin)
    final_tails.append(tails[-1])
plt.hist(final_tails, bins = 10)
plt.show()
```

Histogram, 1,000 runs

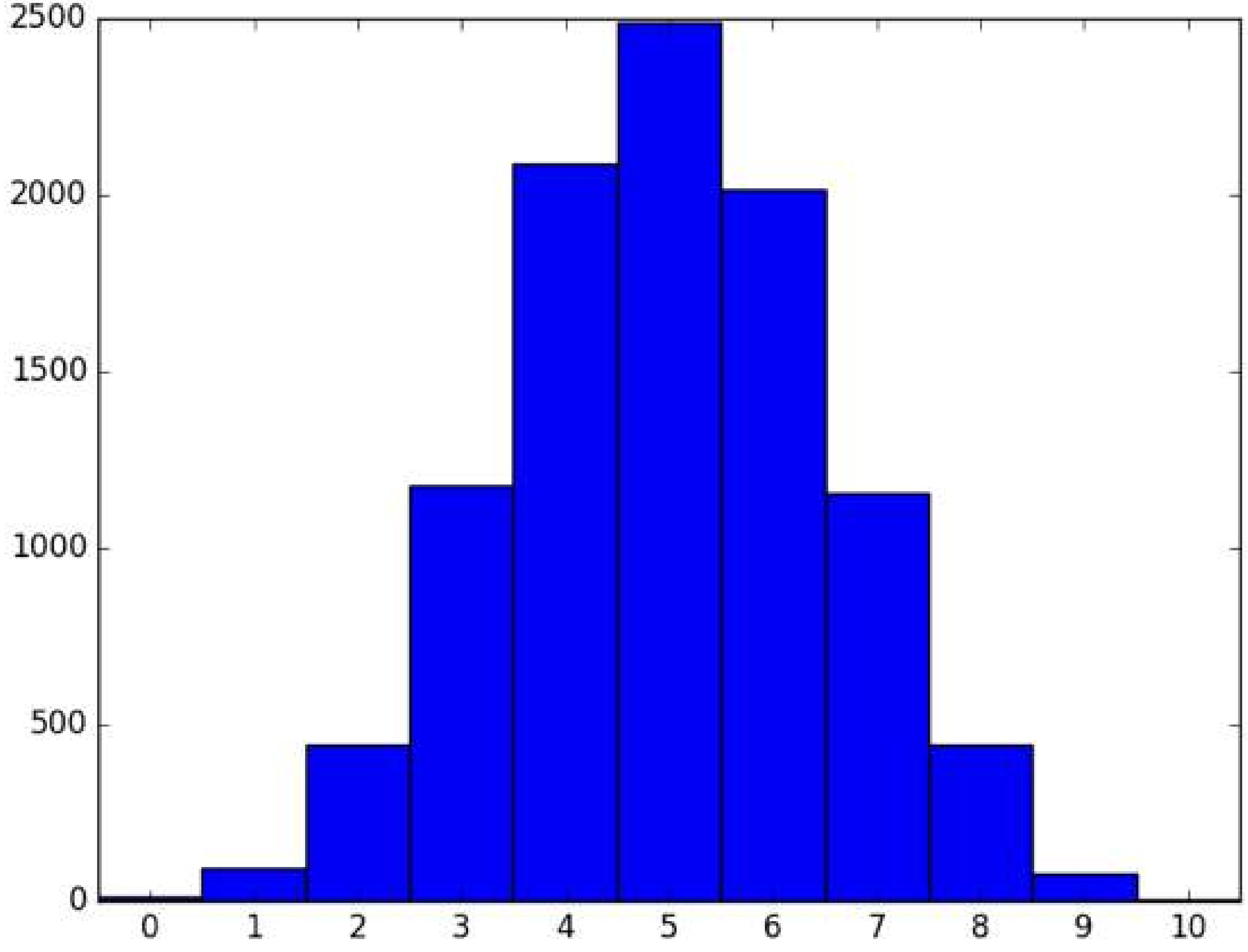


Histogram, 10,000 runs

`distribution.py`

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(123)
final_tails = []
for x in range(10000) : # <--
    tails = [0]
    for x in range(10) :
        coin = np.random.randint(0, 2)
        tails.append(tails[x] + coin)
    final_tails.append(tails[-1])
plt.hist(final_tails, bins = 10)
plt.show()
```

Histogram, 10,000 runs



Let's practice!

INTERMEDIATE PYTHON